

# **XGuardian: Towards Generalized, Explainable and More Effective Server-side Anti-cheat in First-Person Shooter Games**

**Jiayi Zhang, Chenxin Sun, Chenxiong Qian**

***HKU***



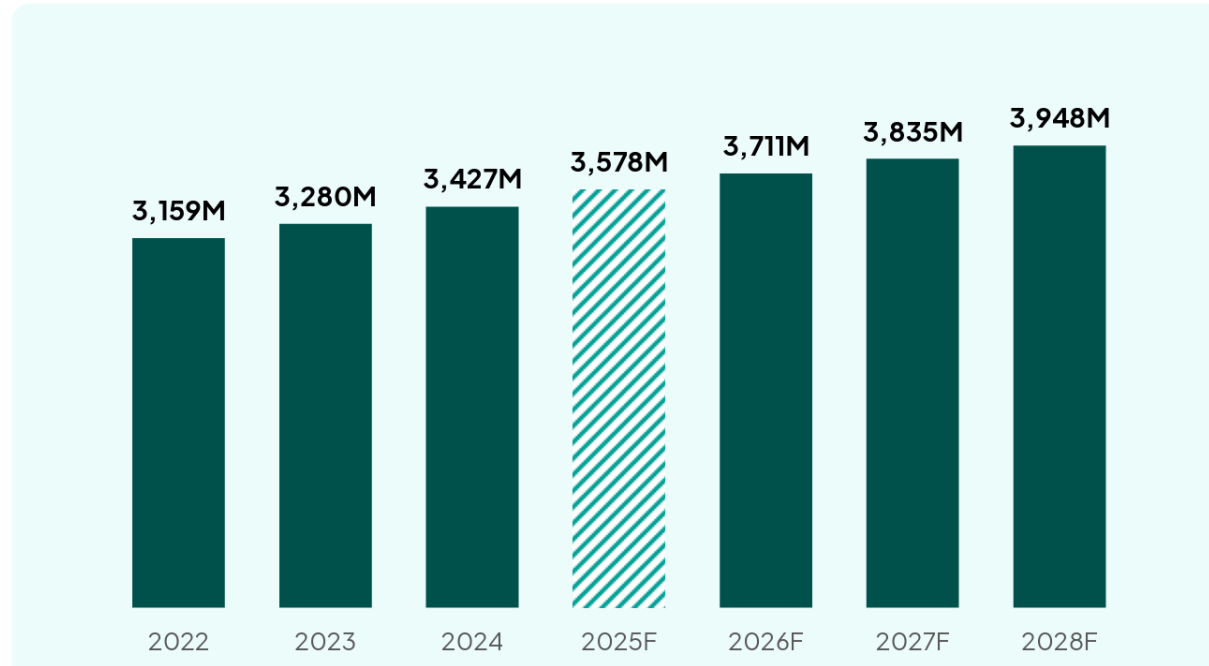
香 港 大 學

THE UNIVERSITY OF HONG KONG

# Continued growth of the gaming industry

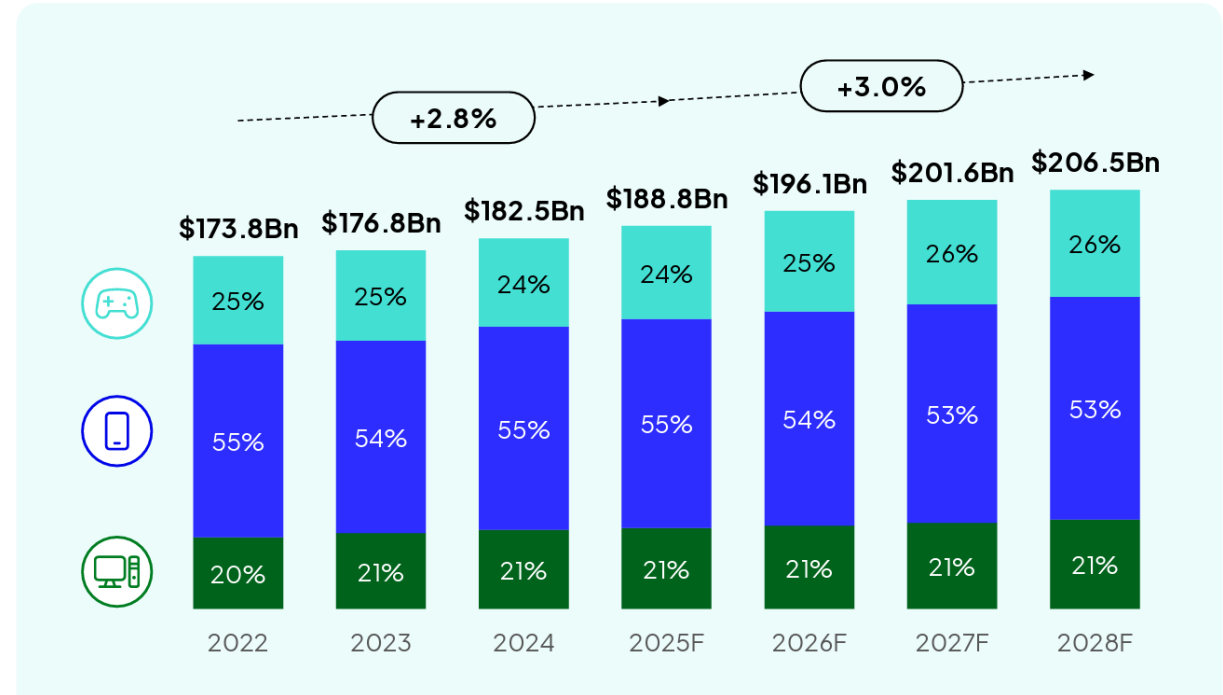
Global player forecast

2022 – 2028F



Global revenue by platform

2022 – 2028F



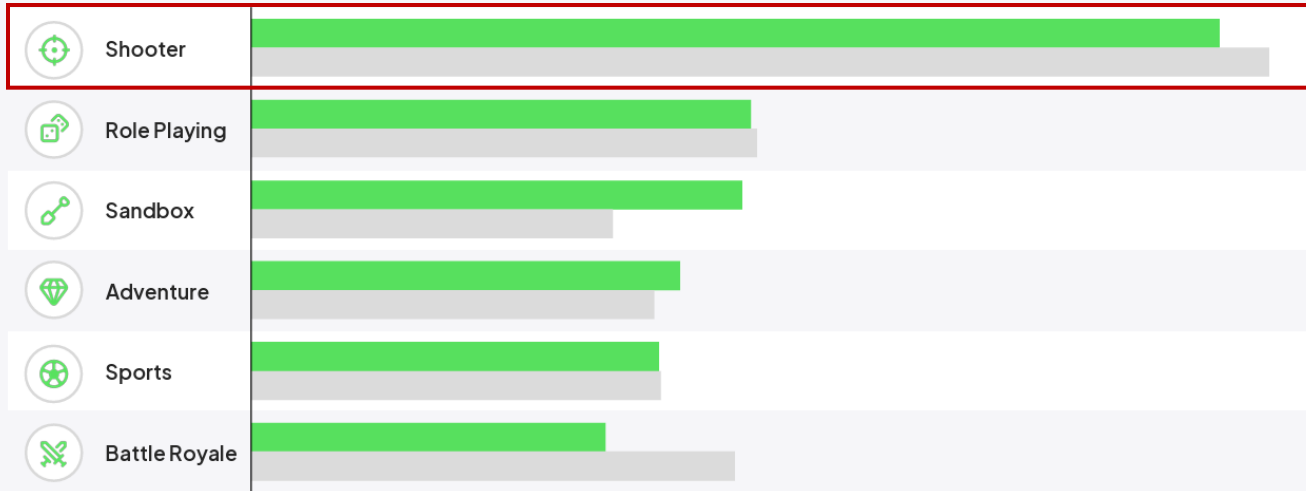
(F indicates forecast, actual statistics for 2025 are still being processed)

# FPS games dominate the market

Top genres by playtime

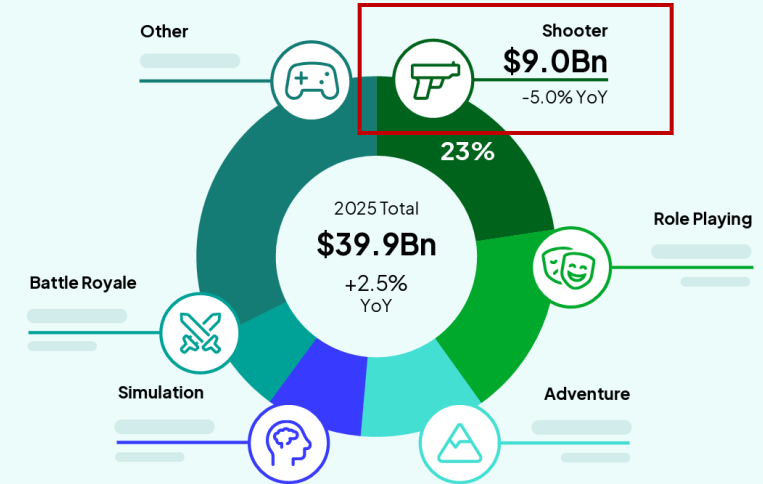
PC, PlayStation, Xbox | 2025 vs. 2024

■ 2025 ■ 2024



Global PC revenues by genre

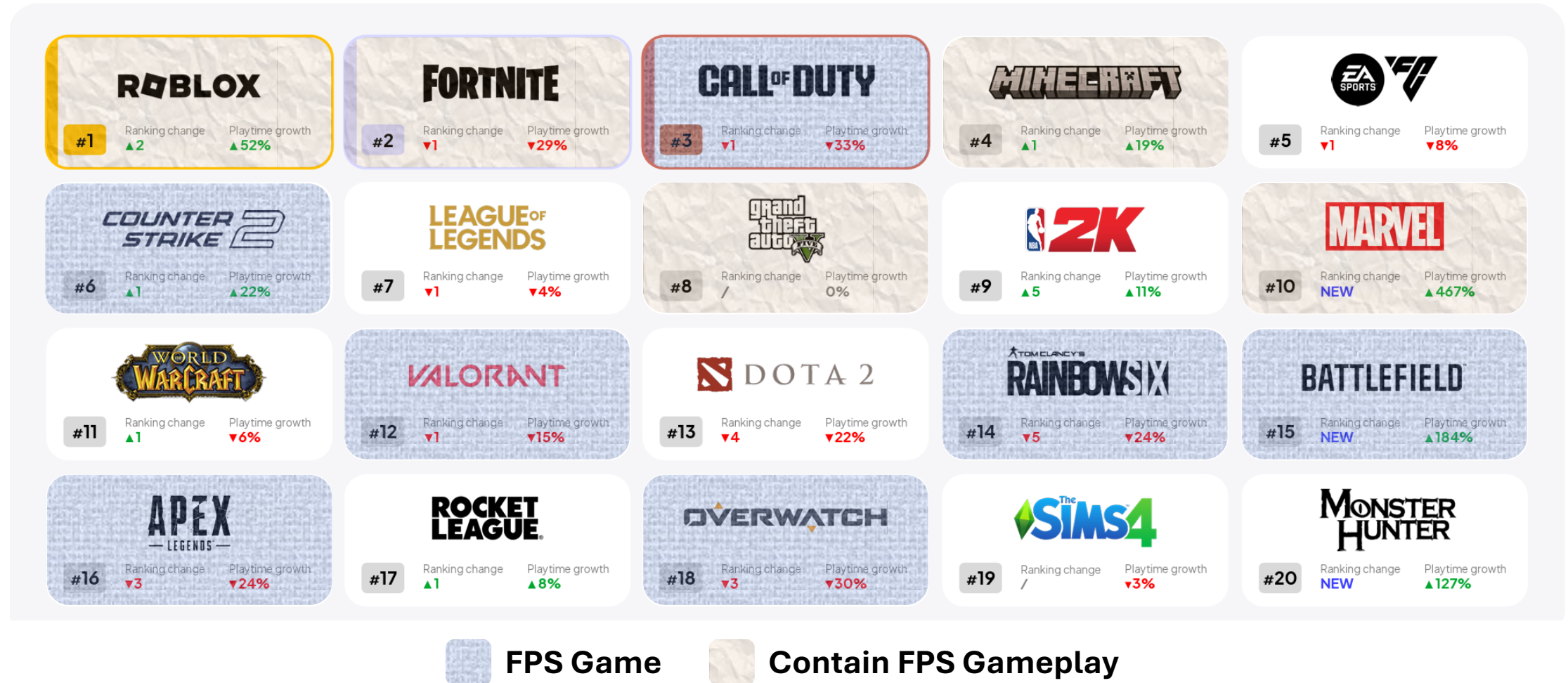
2025F



# FPS games remain a dominant genre

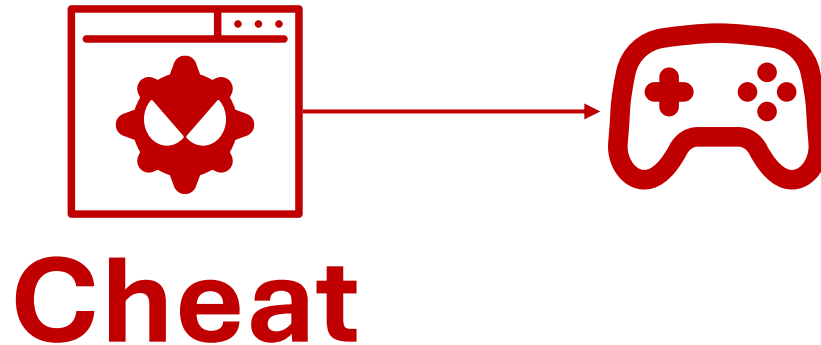
## Top PC/console franchises by overall playtime

PC, PlayStation, Xbox | 2025



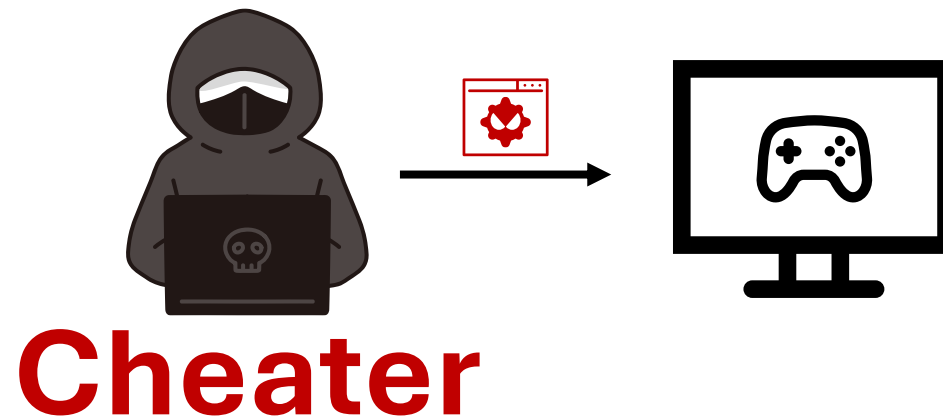
# What is cheat and cheater

- Exploit measures to gain unfair advantages is called the **cheat**
  - usually through malicious software

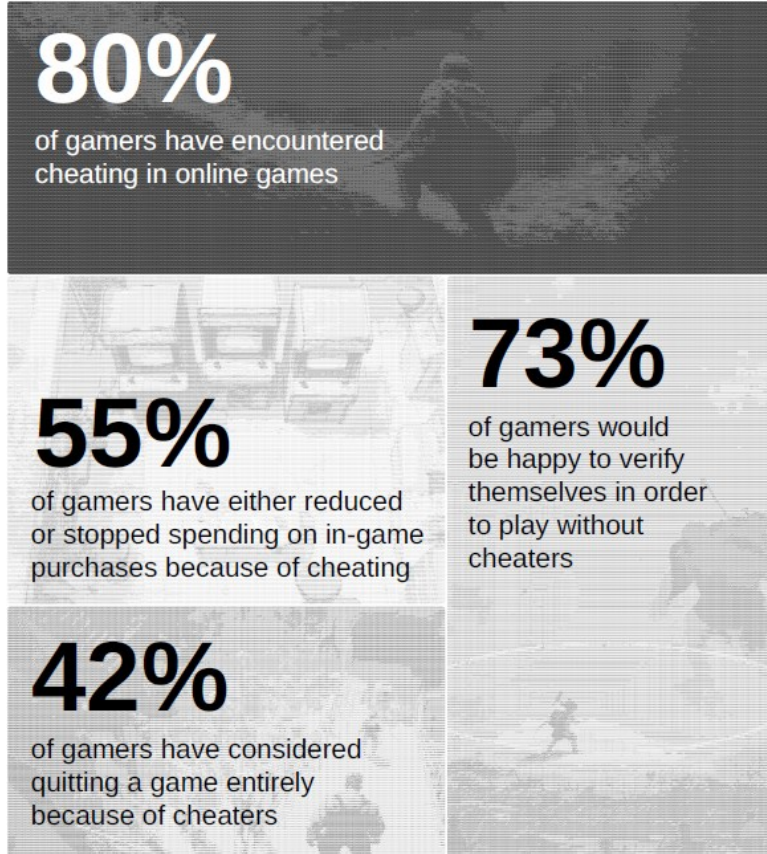


# What is cheat and cheater

- Exploit measures to gain unfair advantages is called the **cheat**
  - usually use malicious software
- The person who uses the cheats is called the **cheater**



# Cheating industry is also “thriving”



# Cheating industry is also “thriving”



China  
(2025)

Cheat Market Size <sup>[1]</sup>  
**\$1.4B – \$1.5B**

NA & EU  
(2025 Incomplete Statistics)

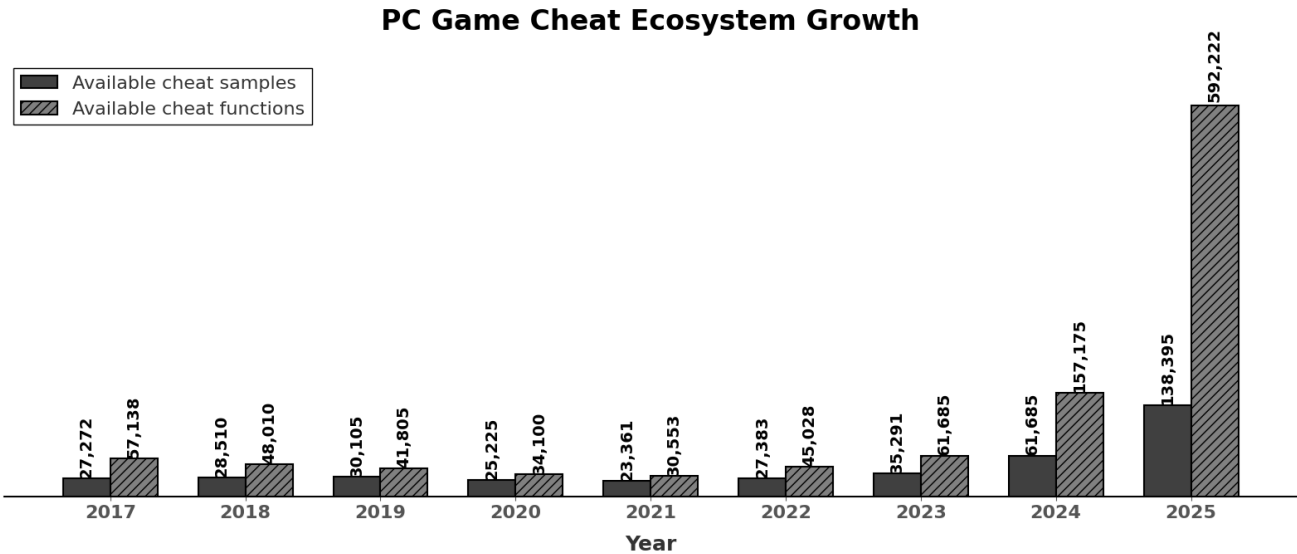
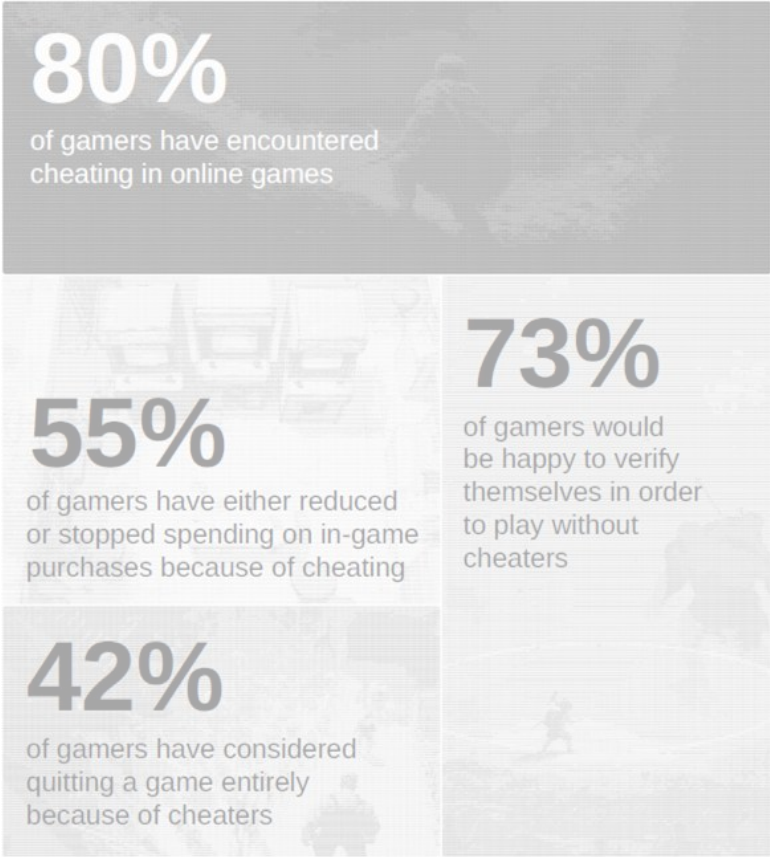
Annual Revenue Per Seller <sup>[2]</sup>  
**\$12.8M – \$73.2M**

[1] Lee Woo Min. (2026, June). *Chinese Game Cheat Market Approaches 2 Trillion Won, Exploding with AI Integration*. <https://www.gamemeca.com/en/view.php?gid=1776690>.

[2] Sam Collins, Alex Pouloupoulos, Marius Muench, and Tom Chothia. 2024. *Anti-Cheat: Attacks and the Effectiveness of Client-Side Defences*. In *Proceedings of the 2024 Workshop on Research on offensive and defensive techniques in the context of Man At The End (MATE) attacks (CheckMATE '24)*. Association for Computing Machinery, New York, NY, USA, 30–43. <https://doi.org/10.1145/3689934.3690816>.



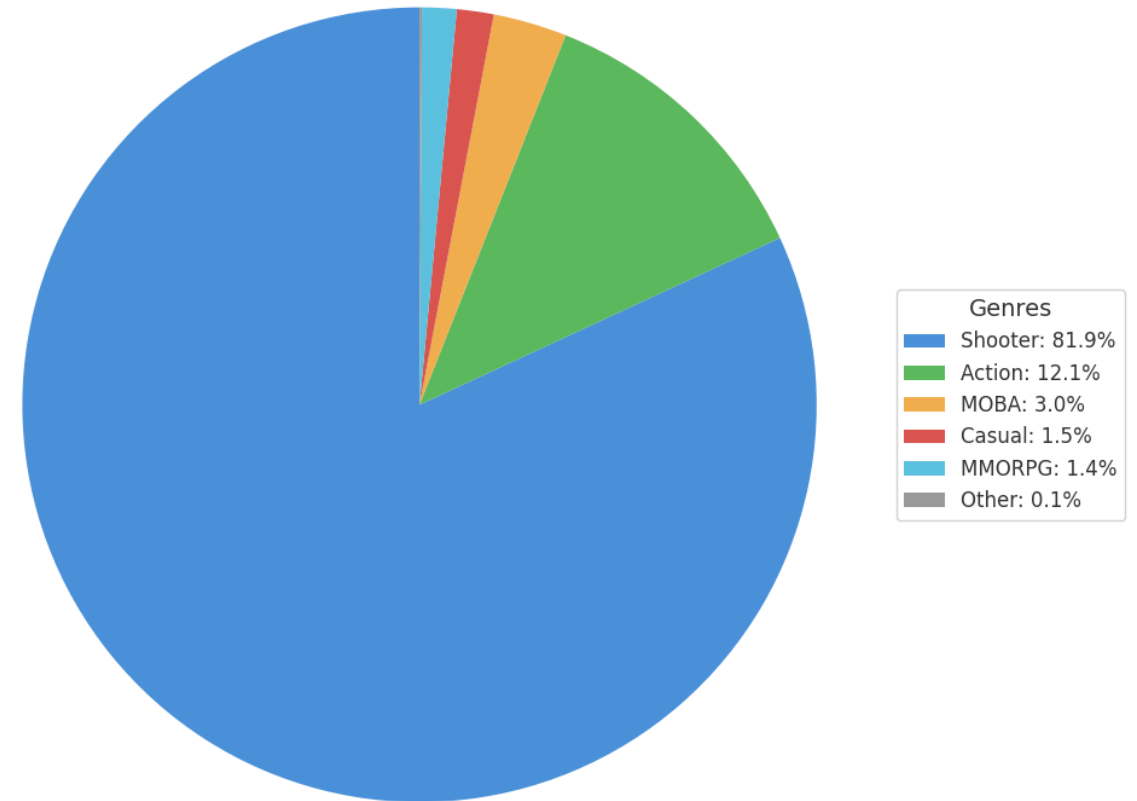
# Cheating industry is also “thriving”



# Cheating is prevalent and harmful in FPS games



2025Q1 Cheater Distribution Across Game Genre



# How do cheaters cheat in FPS games?



## Auto-Aiming Techniques



Starting point



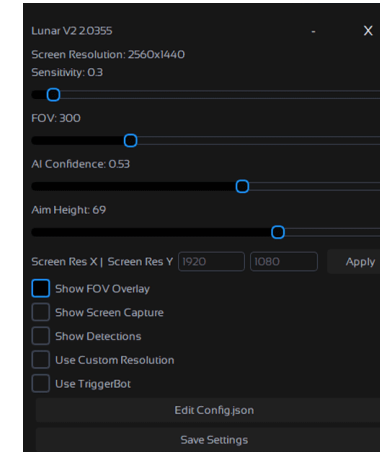
Find the players's health bar



Calculate health bar width.  
Find the red border around  
the player -> calculate body  
coordinates!

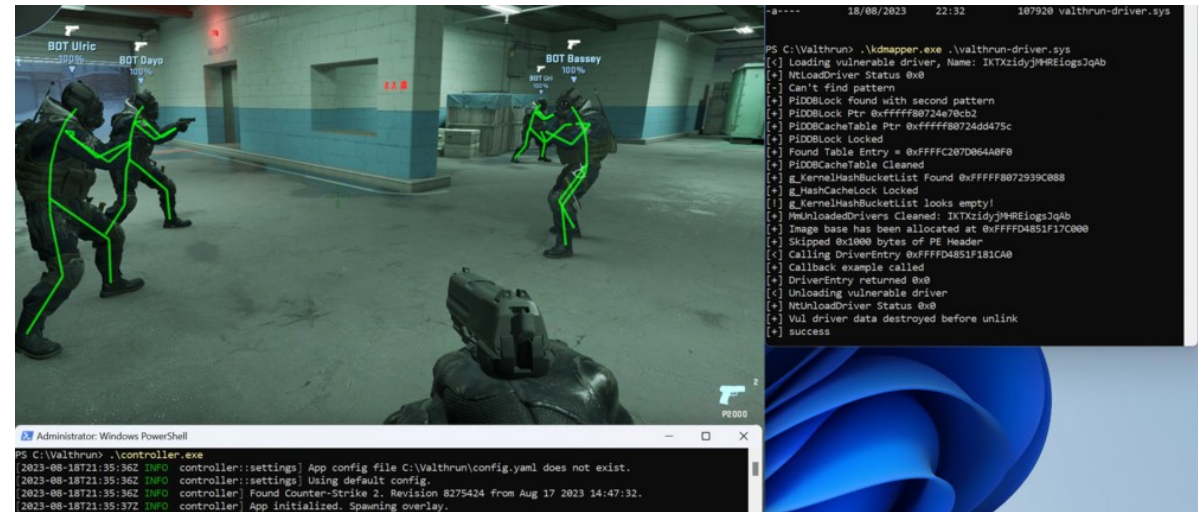
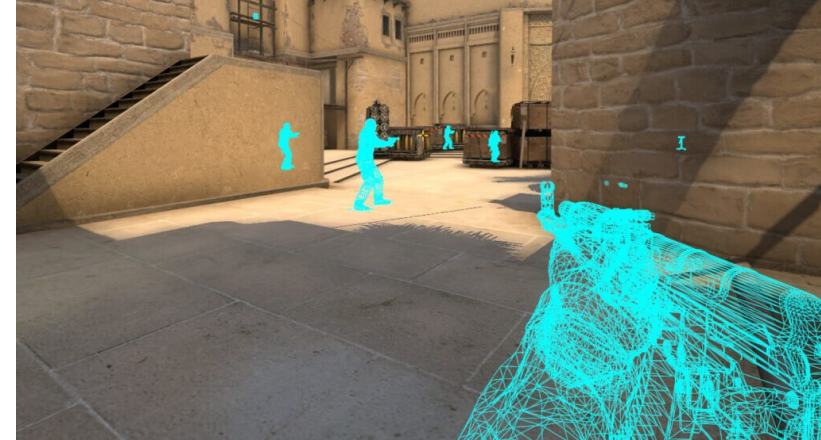
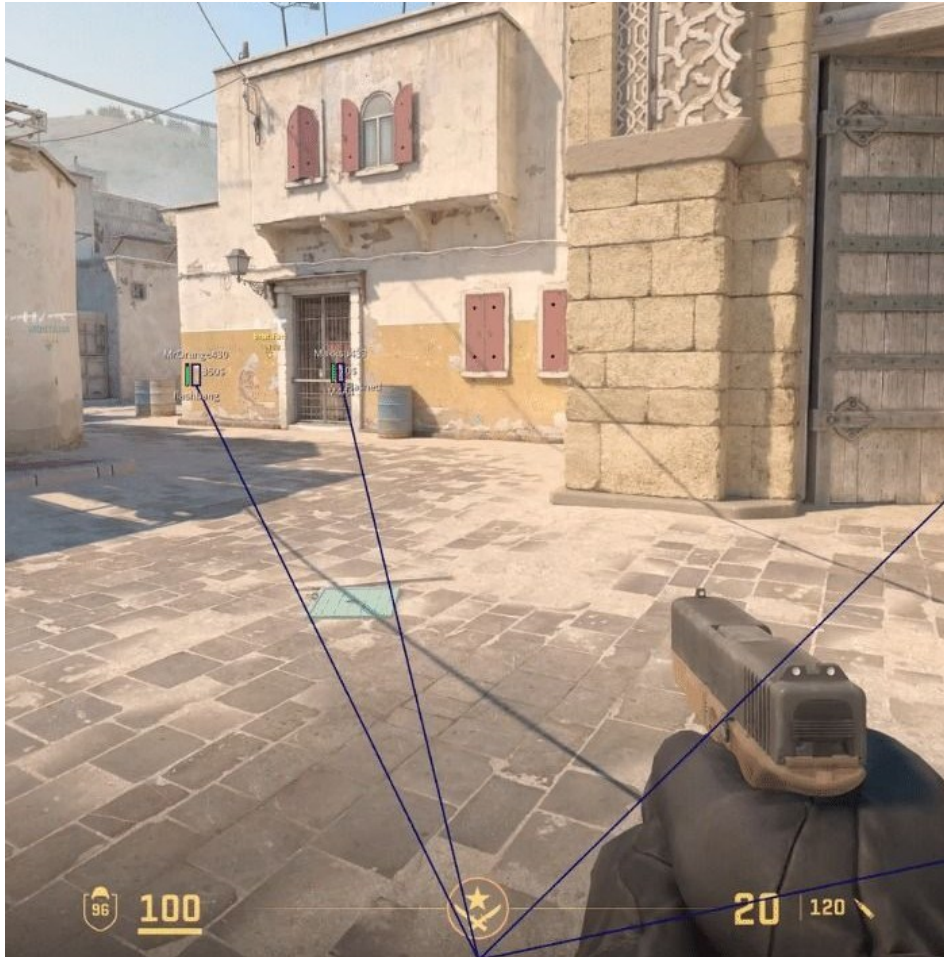


Calculate health bar width.  
Find the red border around  
the player -> calculate head  
coordinates!





# How do cheaters cheat in FPS games?



## Extra-Sensory Perception (ESP) Techniques

# How do cheaters cheat in FPS games?

**Aim-assist** cheats account for the overwhelming majority of FPS cheats

- **TL;DR:** today's cheats are countless, highly integrated and ever-changing
  - ❑ Auto-Aiming techniques
    - e.g., *aimbot (memory, DMA, CV-based), triggerbot, aim trajectory smoothness adjustment...*
  - ❑ ESP techniques
    - e.g., *ESP (skeleton, health, object...), wallhack (bounding box, x-ray, skin highlight...), radar...*
  - ❑ Aim Compensation techniques
    - e.g., recoil control (RCS), no recoil/spread, movement prediction...
  - ❑ Activation techniques
    - e.g., hold key, always on, fire only, toggle...
  - ❑ ...

# How do cheaters cheat in FPS games?

Aim-assist cheats account for the overwhelming majority of FPS cheats

- Highly integrated with hundreds functions including

- ☐ Auto-Aiming techniques

- e.g., *aimbot (memory, DMA, CV-based), triggerbot, aim trajectory smoothness adjustment...*

- ☐ Extra-Sensory

- e.g., *...), radar...*

- ☐ Aim Control

- e.g.,

- ☐ Activation techniques

- e.g., hold key, always on, fire only, toggle...

- ☐ ...

**Different functions combinations**

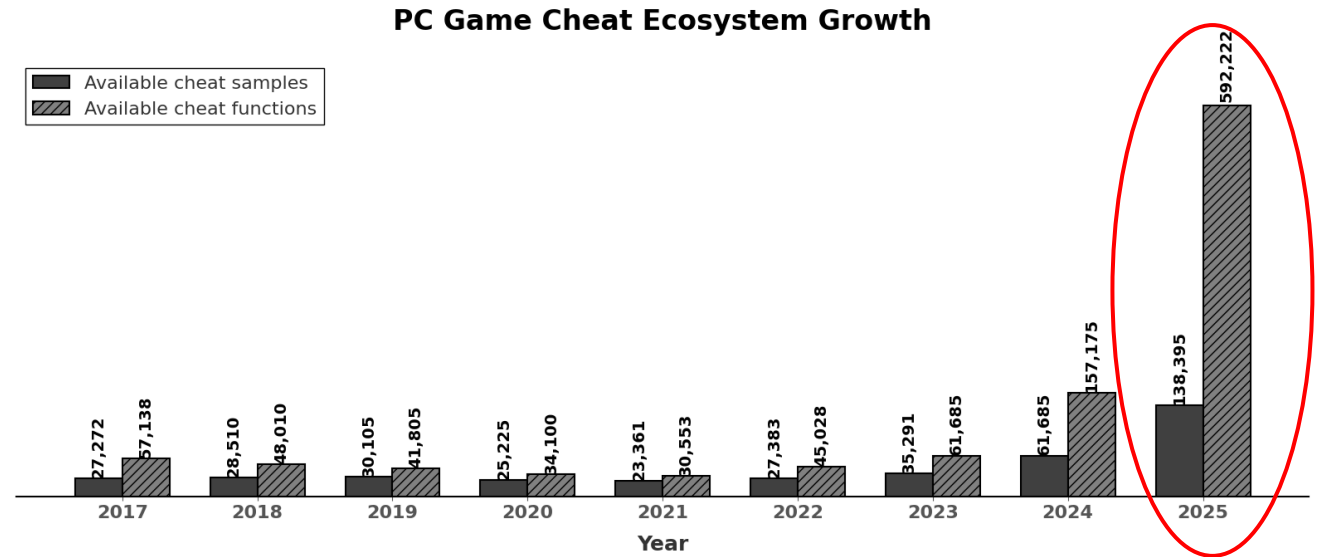
**→ Countless possibilities!**

# Today's real-world FPS anti-cheat up against

***EVERYTHING*** is black-box:

- cheating techniques
- configurations
- functions
- level of cheating sophistication
- cheat developers/vendors

Are all ***DIFFERENT***.



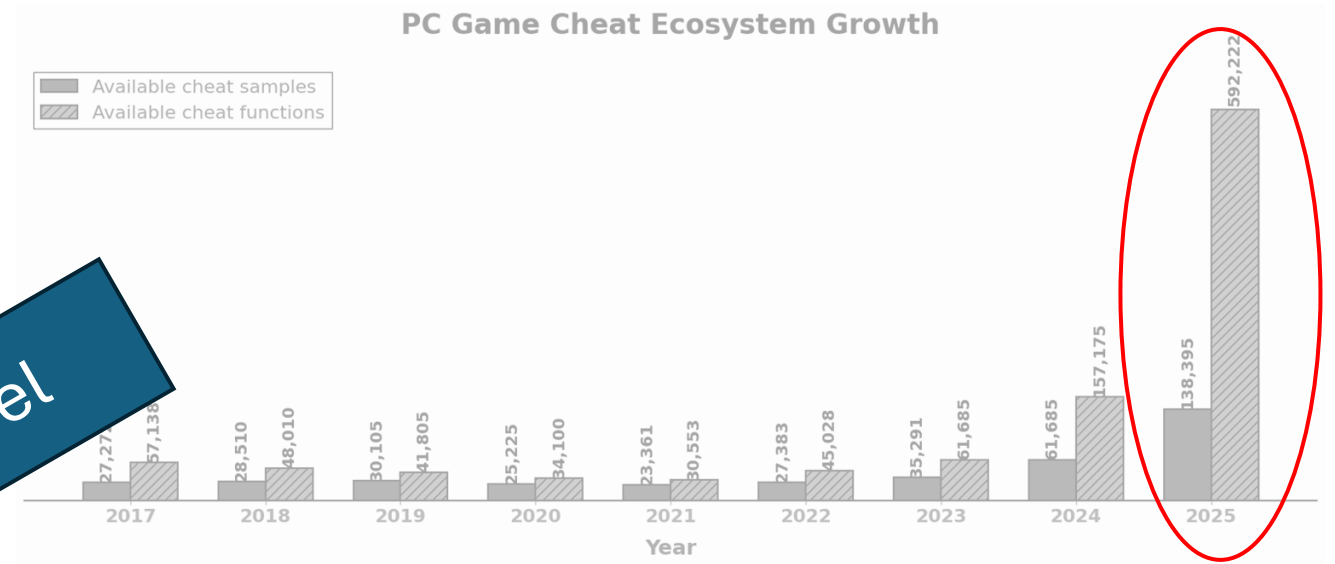
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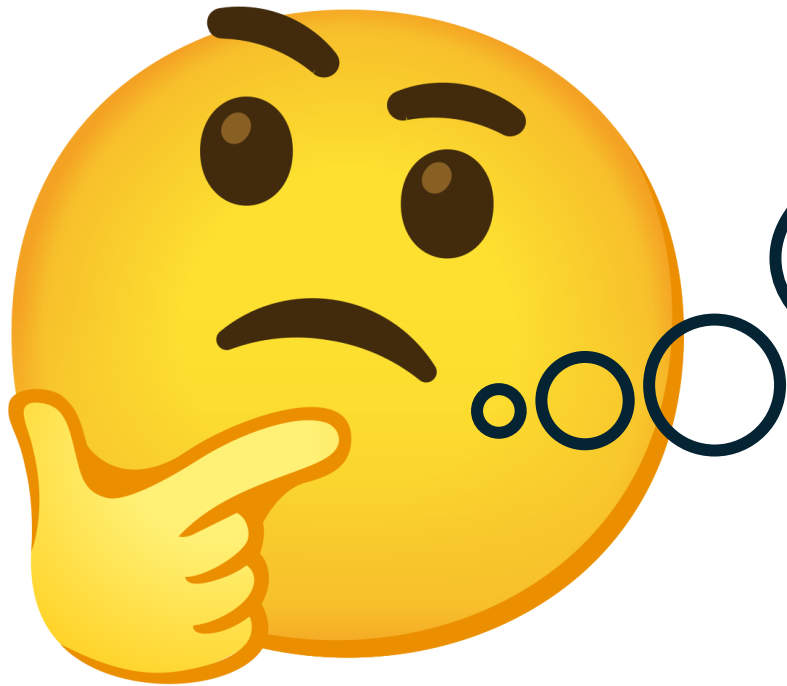
Are all *DIFFERENT*.

Our Threat Model



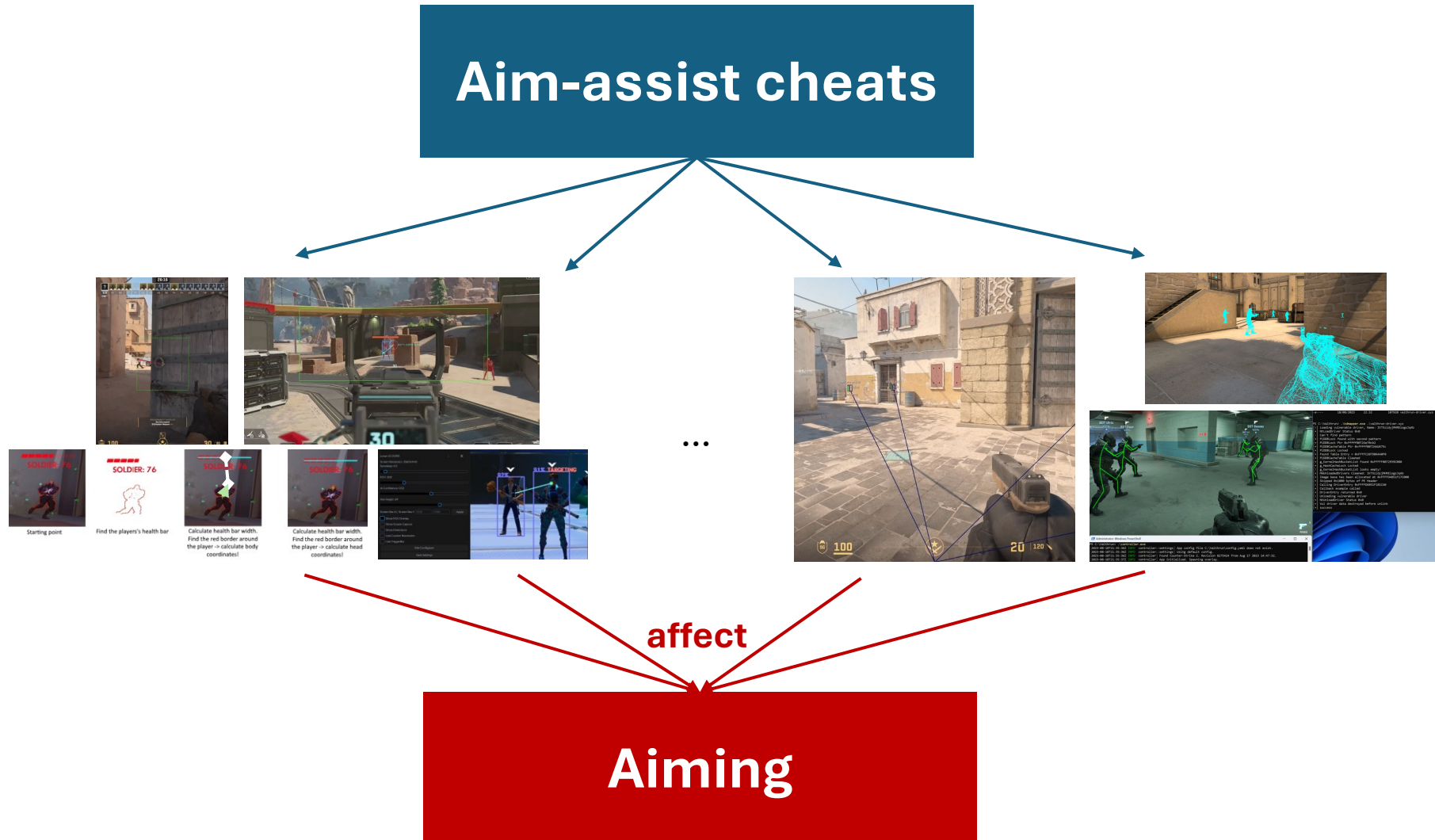


# Is there a silver bullet for aim-assist cheats?

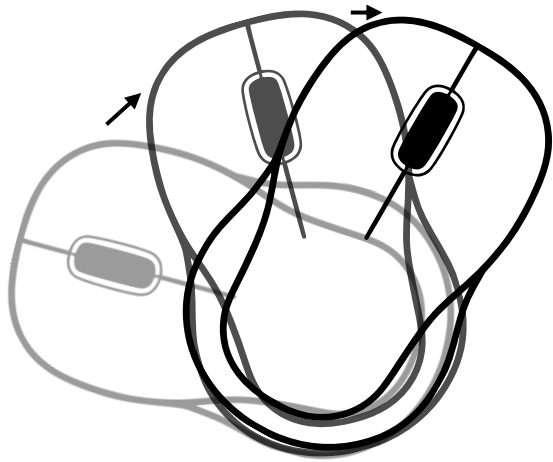


**What do all aim-assist cheats have **in common**?**

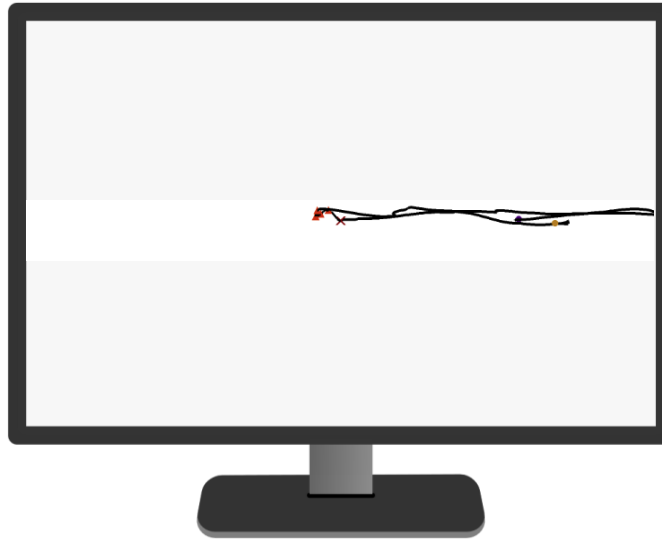
# No aiming, no cheat.



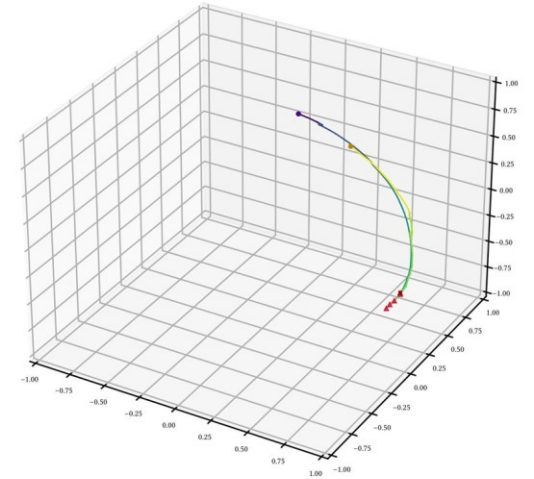
# How does the aiming represent in different systems?



**peripheral moves**



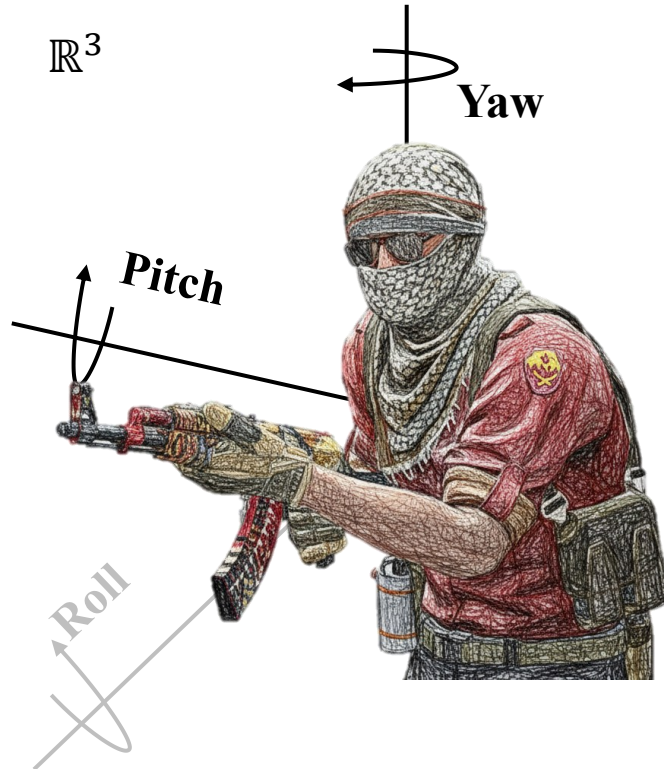
**control crosshair aim on the monitor  
(2D aim trajectory on the screen)**



**control avatar view movement  
(3D aim trajectory in game)**

# How do avatars aim in the game world?

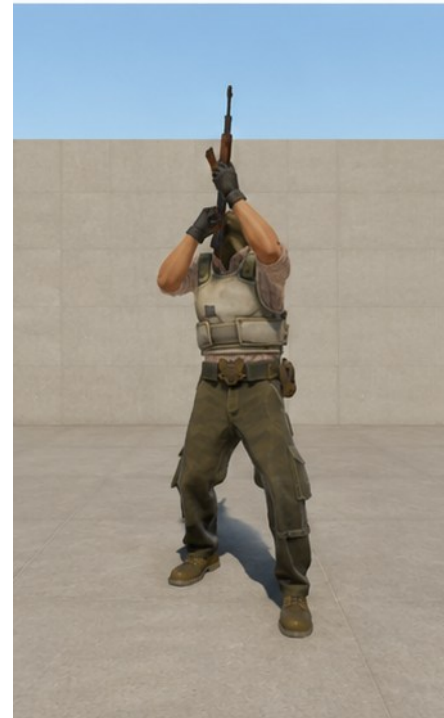
**Pitch** and **yaw** are **must-haves** in FPS games to **control the view**.



# How do avatars aim in the game world?

## Pitch is

- rotating on the **horizontal axis**
- the tilting of the view **up and down**
- range  $[-90^\circ, 90^\circ]$



**$-90^\circ$**   
(looking straight up)

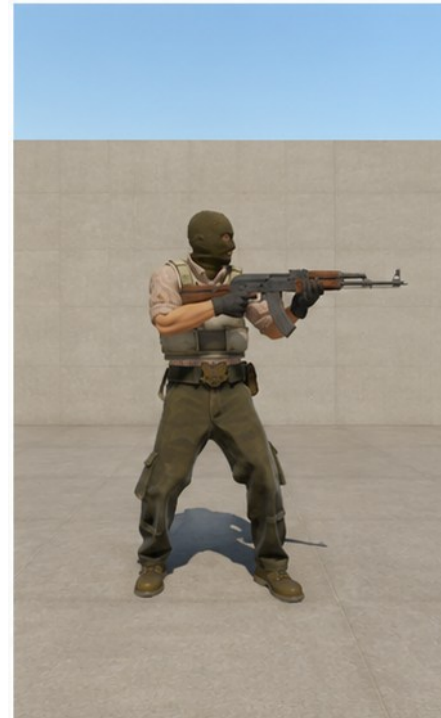


**$90^\circ$**   
(looking straight down)

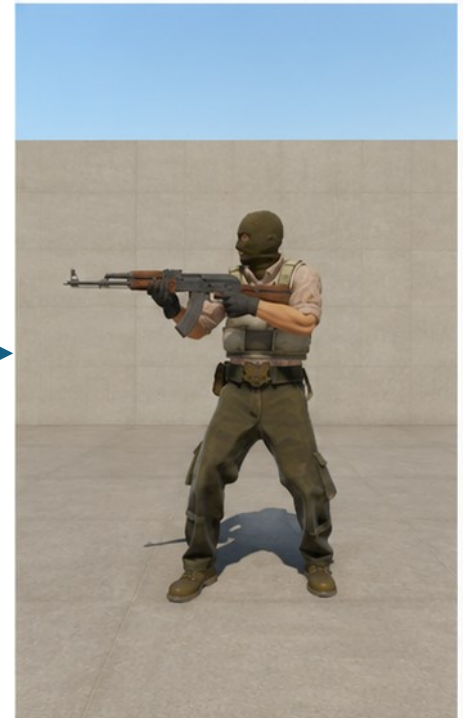
# How do avatars aim in the game world?

## Yaw is

- rotating on the **vertical axis**
- the tilting of the view **left to right**
- range  $[-180^\circ, 180^\circ]$



**$0^\circ$**   
(to the east)



**$-180^\circ$**   
(to the west)



# How can we get the pitch and yaw?

Old fashion



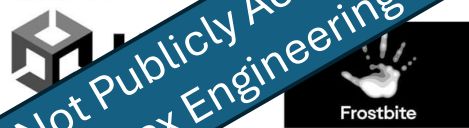
Modify engine



Use cheat's hook

# How can we get the pitch and yaw?

Old fashion



Not Publicly Accessible  
Complex Engineering Effort

Not generalized at all

Modify engine

Use cheat's hook

# How can we get the pitch and yaw?

## Old fashioned



Not Publicly Accessible  
Complex Engineering Effort



Not generalized



Modify engine

Use cheat's hook

## Our method



Replay file

Generalized

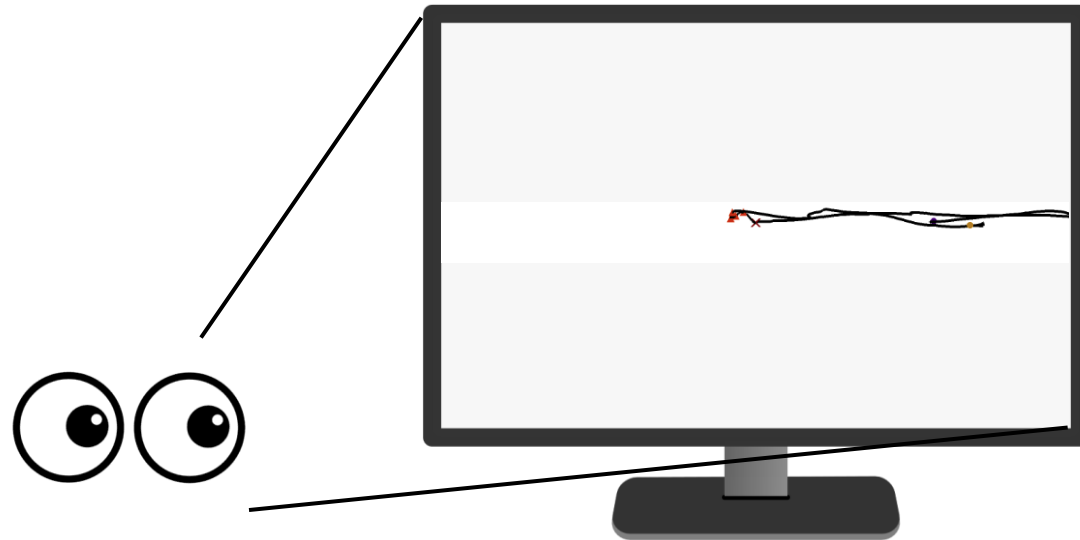
Lossless

Non-Intrusive

Publicly Accessible

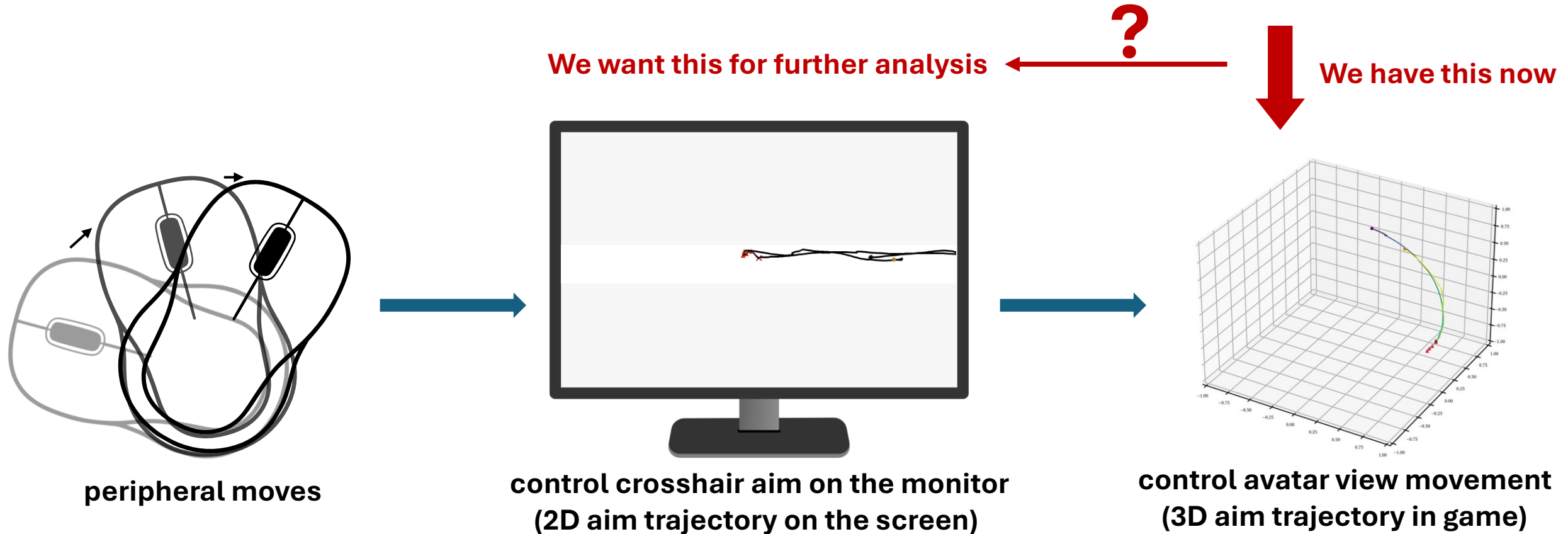
Reconstructible

# Which perspective does AC review expert check?

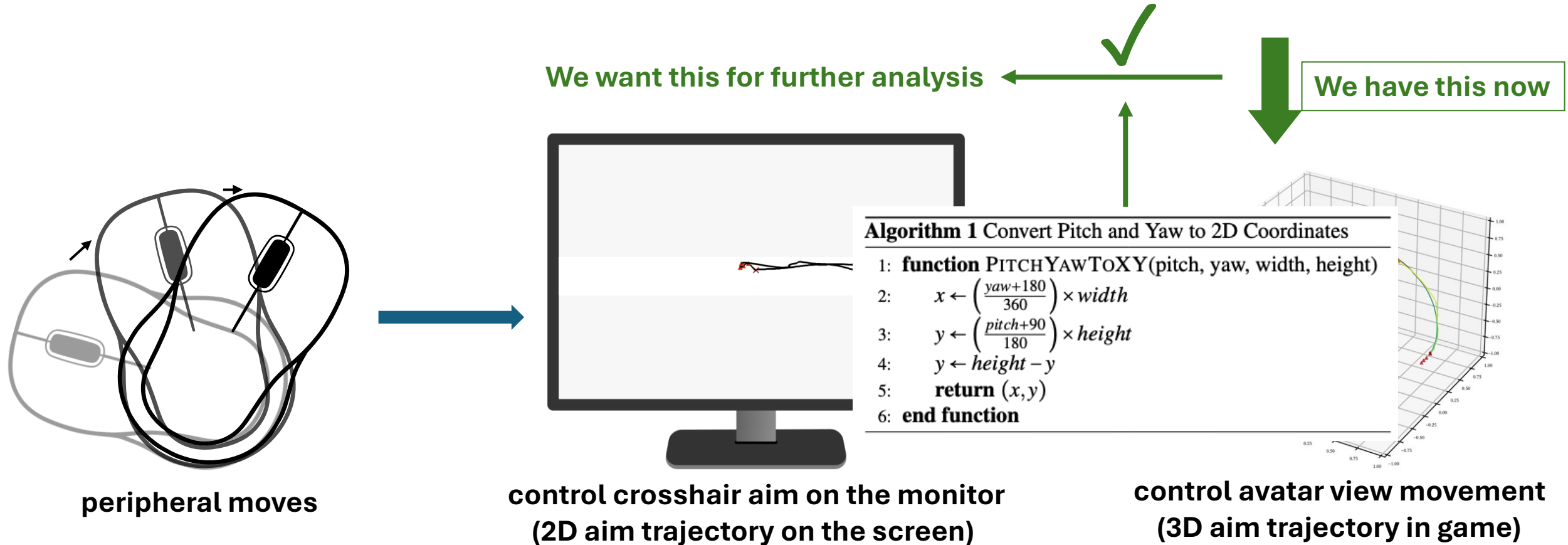


**Watch the suspect's POV  
2D trajectory!**

# How to convert pitch and yaw to 2D trajectory?



# How to convert pitch and yaw to 2D trajectory?





# How does AC review expert identify cheaters?

Check suspect's **POV around kill** and see if there is *unnatural*

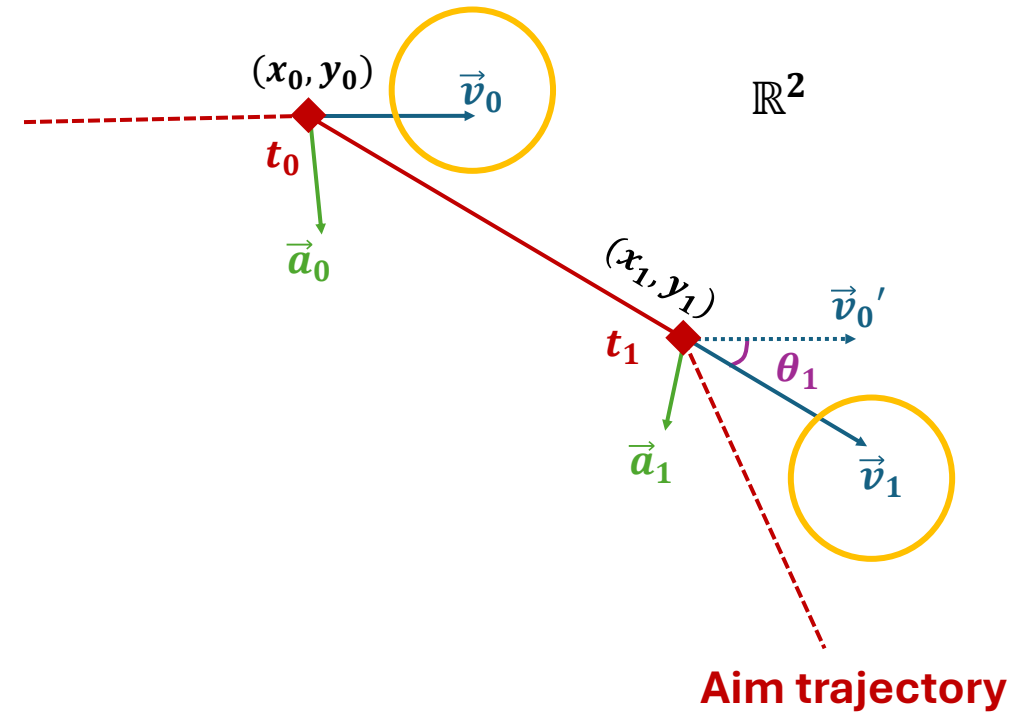
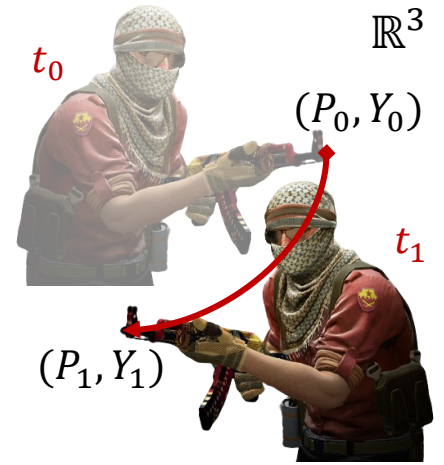
- **aim jitter (sudden bounce)**
- **aim speed and accuracy**
- **pre-aim stableness**

# How to represent these aspects?

Velocities – “aim speed”

$$\vec{v}_x = \frac{dx}{dt}$$

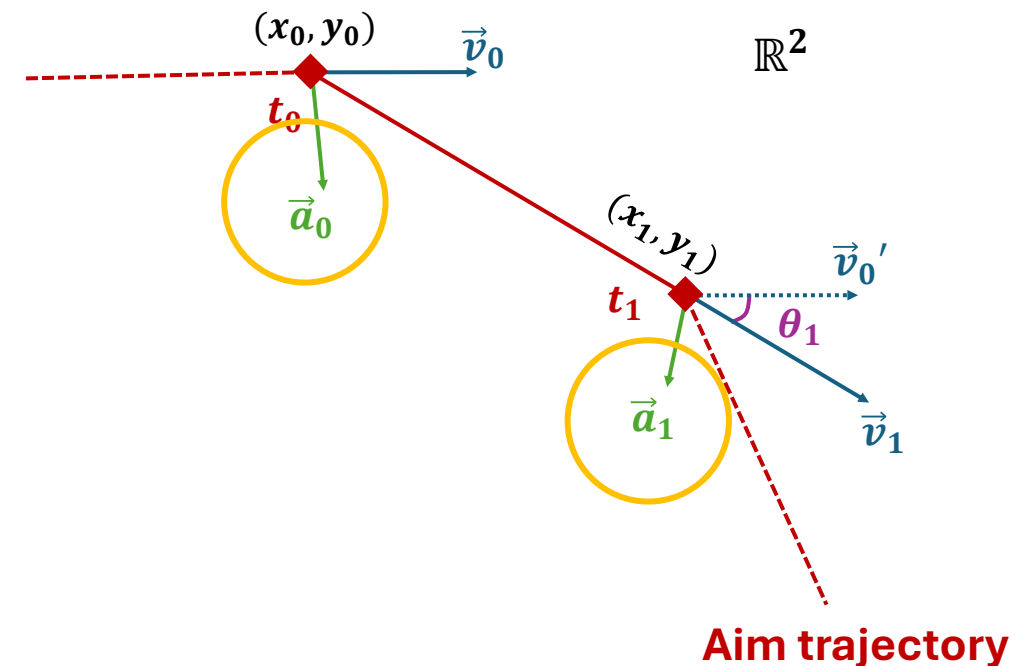
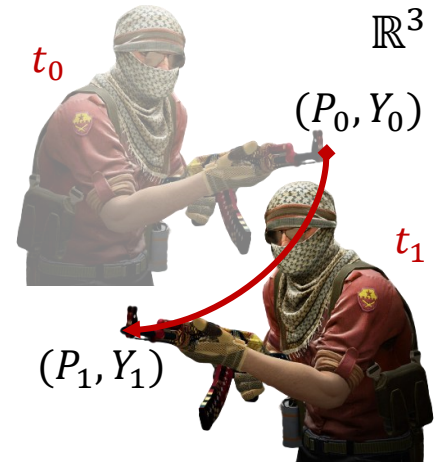
$$\vec{v}_y = \frac{dy}{dt}$$



# How to represent these aspects?

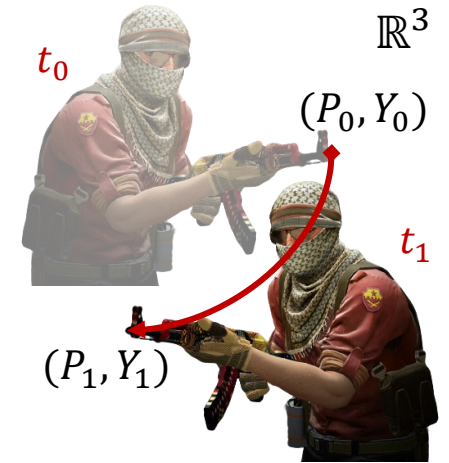
**Accelerations – “aim speed & jitter”**

$$\vec{a}_x = \frac{d\vec{v}_x}{dt} = \frac{d^2x}{dt^2}$$
$$\vec{a}_y = \frac{d\vec{v}_y}{dt} = \frac{d^2y}{dt^2}$$



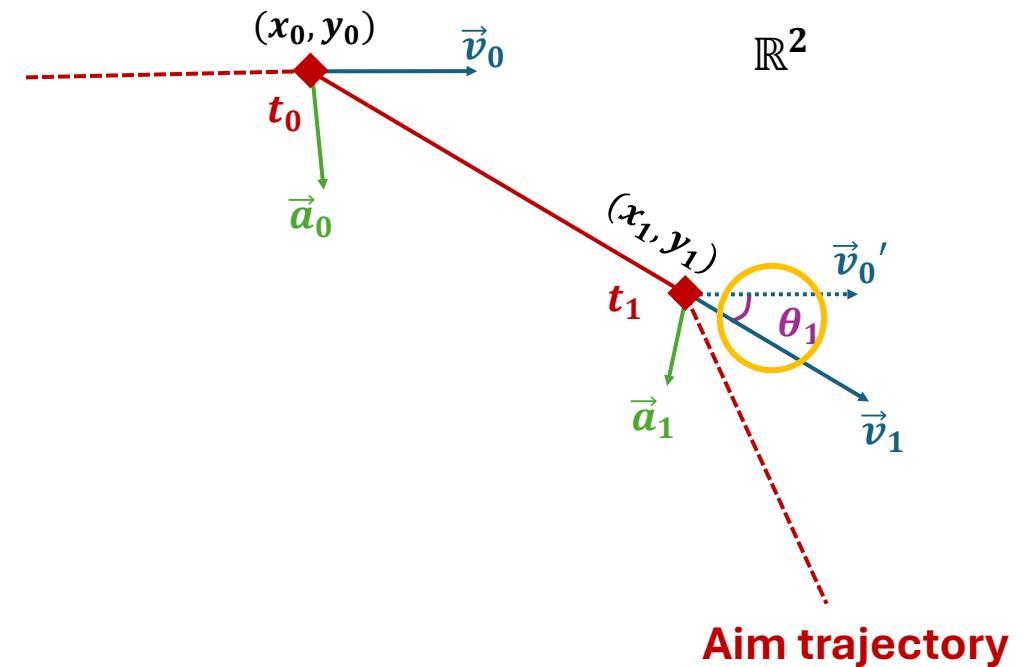
# How to represent these aspects?

Angular change – “aim jitter & pre-aim stableness”



$$\alpha = \arctan 2(\vec{v}_y, \vec{v}_x)$$

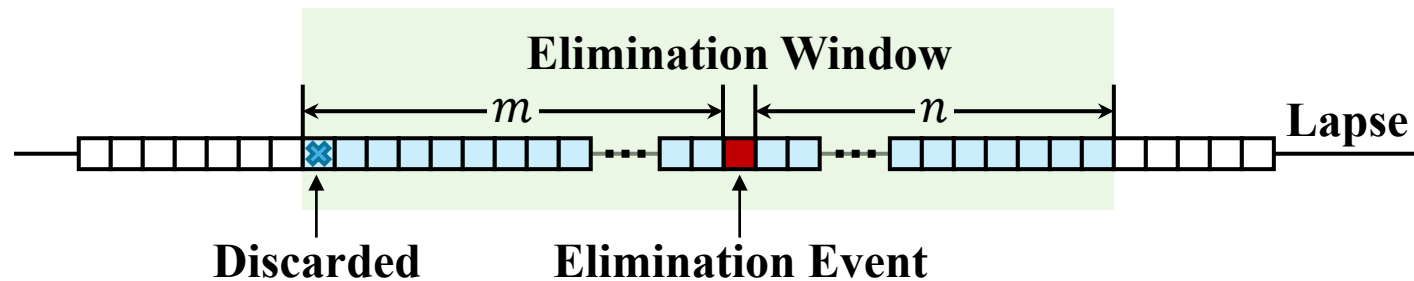
$$\theta = \frac{d\alpha}{dt}$$



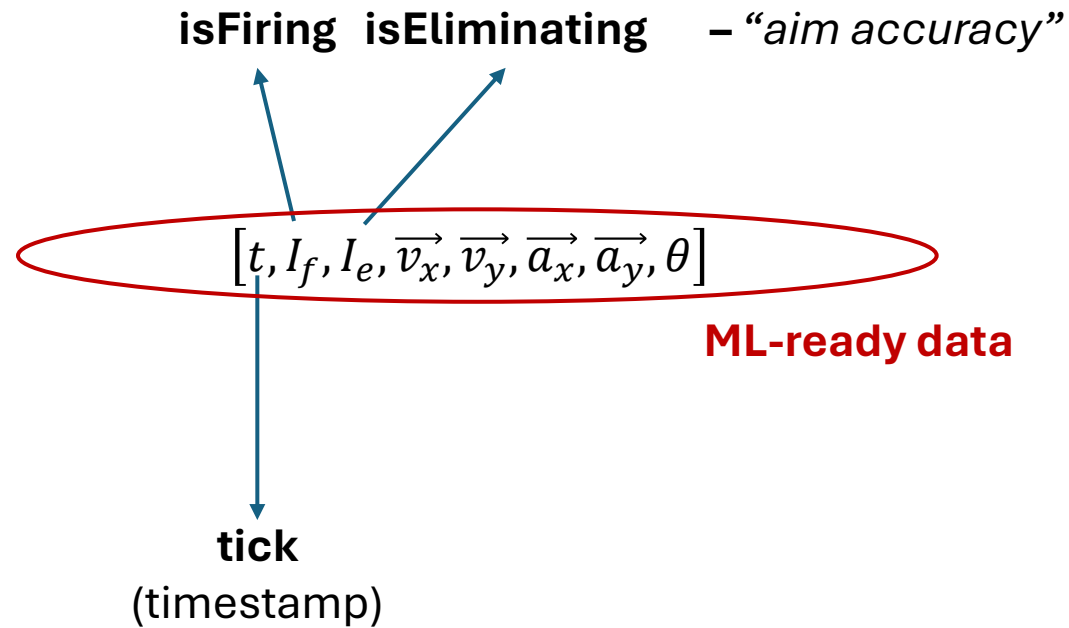
# How to represent these aspects?

## Elimination window – “*around kill*”

- most of the times players are idle – data may introduce **noise**!
- created when a player triggers an elimination event



# How to represent these aspects?



# Implementation

- GRU-CNN + FC aggregation + RF for elimination-wise and match-wise detection
- Modified GradientSHAP + TreeSHAP to explain the predictions



# XGuardian contributions

- Effective, efficient and generalized solution
- Explainable visualizations for each kill and each match
- Real-world datasets

# Datasets

## Primary evaluations

- *Counter Strike 2*
- Real-world data and matches from *5E-Play* [1]

Table 2: Dataset description.

| Data Split                   | A       | B         | C         |
|------------------------------|---------|-----------|-----------|
| #Tick                        | 992,064 | 1,014,624 | 1,062,528 |
| #Total Elimination Window    | 10,334  | 10,569    | 11,068    |
| #Cheating Elimination Window | 2,385   | 2,760     | 2,900     |
| #Normal Elimination Window   | 7,949   | 7,809     | 8,168     |
| #Match                       | 971     | 944       | 988       |
| #Cheater                     | 108     | 115       | 127       |
| #Normal Player               | 1,711   | 1,671     | 1,754     |
| #Total Player                | 1,819   | 1,786     | 1,881     |

# Datasets

## Generalizability evaluation

- *Farlight84* & *CS:GO* (from *HAWK* [1])
- Real-world data and matches from *5E-Play* and *Lilith Games* [2]

Table 18: FARLIGHT84 dataset descriptions.

| Scale         | Train   |          | Validation |          | Test    |          |
|---------------|---------|----------|------------|----------|---------|----------|
|               | #Normal | #Cheater | #Normal    | #Cheater | #Normal | #Cheater |
| <b>Sample</b> | 11,402  | 2,601    | 8,552      | 1,951    | 8,553   | 1,952    |
| <b>Match</b>  | 7,554   | 1,493    | 6,147      | 1,232    | 6,149   | 1,256    |
| <b>Player</b> | 5,513   | 64       | 4,678      | 64       | 4,691   | 66       |

Table 19: HAWK dataset descriptions.

| Scale         | Train   |          | Validation |          | Test    |          |
|---------------|---------|----------|------------|----------|---------|----------|
|               | #Normal | #Cheater | #Normal    | #Cheater | #Normal | #Cheater |
| <b>Sample</b> | 152,113 | 12,273   | 146,543    | 11,970   | 146,887 | 12,313   |
| <b>Match</b>  | 1,007   | 539      | 975        | 528      | 993     | 541      |
| <b>Player</b> | 8,971   | 357      | 8,660      | 336      | 8,800   | 336      |

# What experiments have we done?

- Detection performance
- Feature ablation study
- Inspector's model and parameter selection
- Comparative study with prior works
- Generalizability study
- Explainability case study
- Robustness study under adversarial attack
- Overhead study
- User study on Explainer's potential bias

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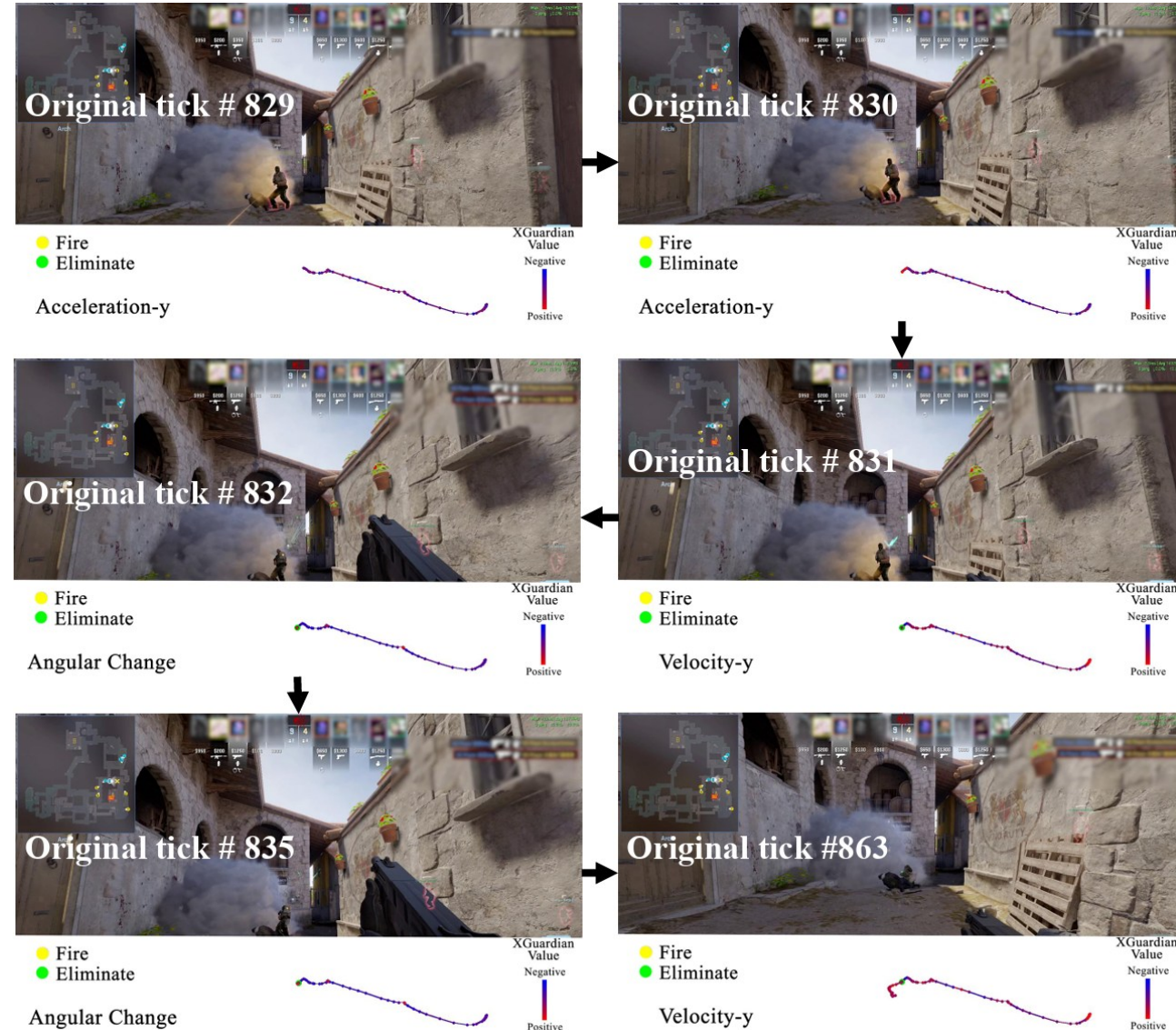
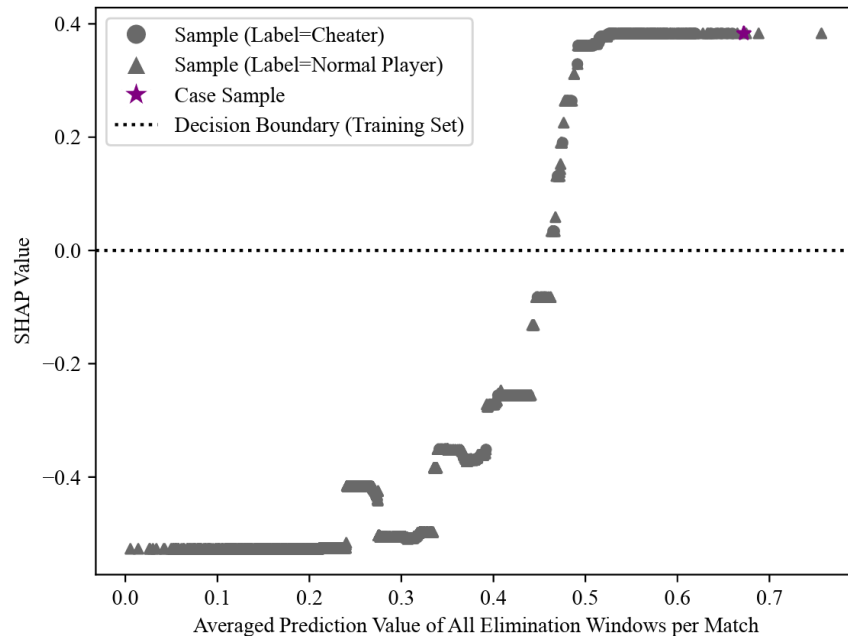
# Generalizability study

- Different titles, sub-genres, platforms and game engines
- Stable performance
- Results are weighted averaged due to imbalanced dataset

| Dataset             | HAWK [63]        | FARLIGHT84        | XGUARDIAN        |
|---------------------|------------------|-------------------|------------------|
| Game (Release Year) | CS:GO (2012)     | Farlight84 (2023) | CS2 (2023)       |
| Sub-genre           | Tactical Shooter | Battle Royal      | Tactical Shooter |
| Engine              | Source           | Unreal Engine 4   | Source 2         |
| Platform            | PC               | Mobile            | PC               |
| Accuracy            | 0.841            | 0.816             | 0.889            |
| FPR                 | 0.056            | 0.144             | 0.062            |
| Recall              | 0.841            | 0.816             | 0.889            |
| Precision           | 0.944            | 0.856             | 0.938            |
| F1-Score            | 0.878            | 0.829             | 0.907            |

# Explainability case study

- **Per-frame** visualization analysis for different features
- **Per-match** analysis for demonstrating the distribution among all matches



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## Links

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Artifact



Paper & Tech Rec.



Demo



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香 港 大 學

THE UNIVERSITY OF HONG KONG

# Gaming industry is thriving

Online gaming market is projected to grow to **\$386 billion** by 2026.



Picture Sources:

[1] [Liquipedia Wiki](#)

[2] [OWCS EWC 25 Champions](#)

# FPS game genre is market leader

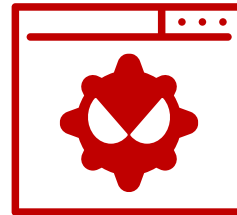
First-Person Shooter (**FPS**) game holds 20.9% market share.



Video Source:  
[\[1\] Counter-Strike 2](#)

# FPS games are vulnerable to cheating

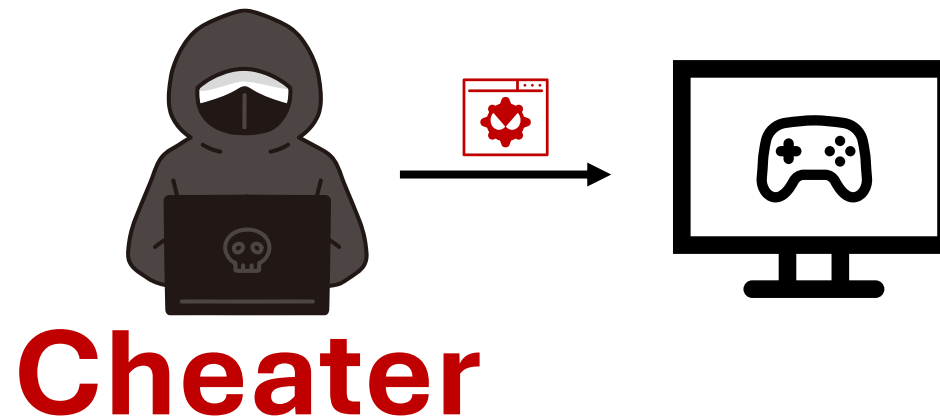
- Malicious software for games is called the **cheat**
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## Cheat

# FPS games are vulnerable to cheating

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  - Cheat developers seek for monetary gain
- The person who uses the cheats is called the **cheater**
  - cheaters seek in-game illegal advantages against their opponents



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- Malicious software for games is called the **cheat**
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- **Cheating** poses user churn and financial loss

# FPS games are vulnerable to cheating

- Malicious software for games is called the **cheat**
  - Cheat developers seek for monetary gain
- The person who uses the cheats is called the **cheater**
  - cheaters seek in-game illegal advantages against their opponents
- **Cheating** poses user churn and financial loss
- **Anti-cheat** aims to detect or prevent cheating



# Aim-assist cheats in FPS Games

Two types of aim-assist cheats in modern FPS games:

- **Aimbot** – auto-aim and auto-shoot





# Aim-assist cheats in FPS Games

Two types of aim-assist cheats in modern FPS games:

- **Aimbot** – auto-aim and auto-shoot
  - Mutations: triggerbot (left), ai-aimbot (right)

Video Source:  
[\[1\] aptitude.pub](#)



# Aim-assist cheats in FPS Games

Two types of cheats in modern FPS games:

- **Wallhack** – see through obstacles



# Aim-assist cheats in FPS Games

Aim-assist cheats commonly integrate *aimbots* with *wallhacks* and constitute a significant portion of all FPS cheats.

**XGuardian addresses aim-assist cheats.**

# Server-side and client-side anti-cheat

Server-side anti-cheat systems are defined by their architectural design to collect, process, and analyze gameplay data exclusively on secure server infrastructure.

# Existing solutions' limitations

**[L1]** Client-side dependency and resource constraints

**[L2]** Limited real-world performance

**[L3]** Limited generalizability

**[L4]** Lack of explainability

# Existing solutions' limitations

[L1] Client-side dependency and resource constraints

[L2] Limited real-world performance

[L3] Limited generalizability

[L4] Lack of explainability

Table 1: ML-based anti-cheat detection works comparison.

| Side   | Method                                 | Work(s)              | Availability <sup>1</sup> | Generalizability |                      |                        | Interpretability | Real-world Functionality |                     |                      |
|--------|----------------------------------------|----------------------|---------------------------|------------------|----------------------|------------------------|------------------|--------------------------|---------------------|----------------------|
|        |                                        |                      |                           | Data Collection  | Feature <sup>2</sup> | Evaluated <sup>3</sup> |                  | Dataset                  | System Constraint   | Performance          |
| Client | Unsupervised Angle Anomaly Detection   | [6] <b>BotScreen</b> | <b>T, C, D</b>            | ×                | ✗                    | ×                      | ×                | SSL <sup>4</sup>         | TEE (Intel SGX)     | Unknown <sup>5</sup> |
|        | Statistical Classification             | [8, 11]              | <b>T, C, D</b>            | ×                | ✓                    | ×                      | ×                | SSL                      | Concurrent overhead | Unknown              |
| Server | Statistical Classification             | [7, 9, 10]           | <b>T, C, D</b>            | ×                | ✓                    | ×                      | ×                | SSL                      | Concurrent overhead | Unknown              |
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## Explainable **Guardian** (XGuardian)

- Avoids the inherent limitations of client-side solutions and leverages replay files as the data source to perform detection asynchronously during server idle periods, minimizing overhead [L1].

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- Addresses the explainability bottleneck by instantly pinpointing the exact suspicious frames and features within a match [L4].



# Replay system and replay file

- **Replay system** is a dynamic toolset for player interaction with recorded content post-game and has been extensively adopted in notable FPS games
- **Replay file** is a comprehensive record of a match with unique data format that can be interpreted by the replay system

**XGuardian uses replay files to extract raw data.**

# Cheating sophistication

Cheaters dynamically adjust sophistication levels of using aim-assist cheats across two axes:

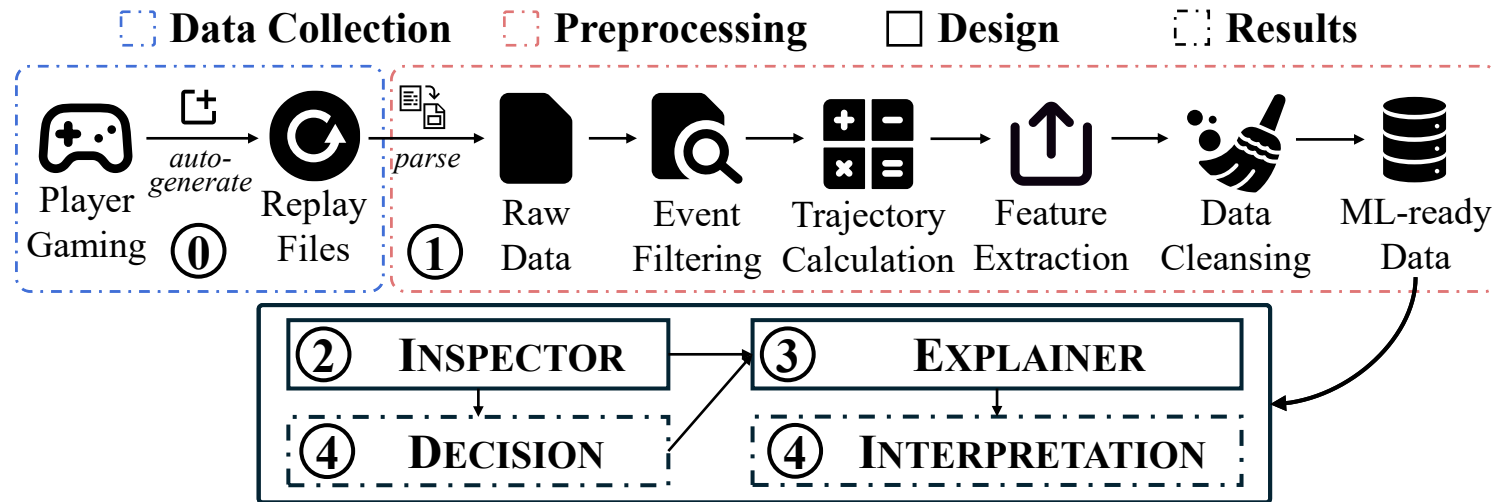
- Functional granularity (e.g., targeting specific body part vs. full-body locks)
- Stealth tradeoffs (e.g., blatant play vs. behavioral mimicry of legitimate players)

# Threat Model

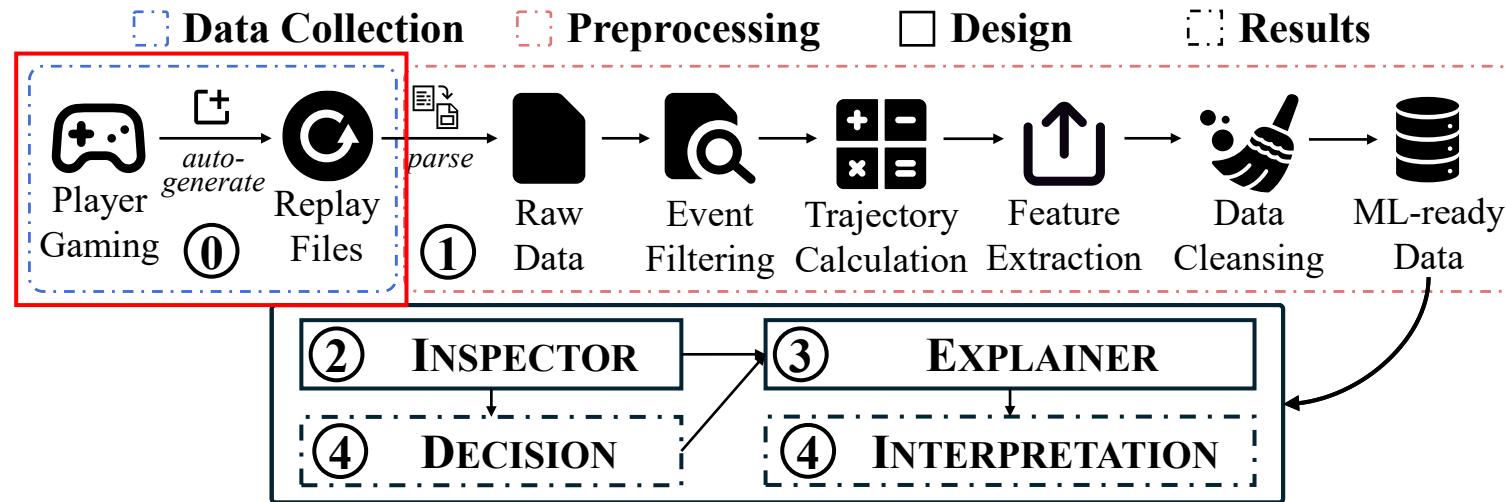
## The cheaters

- Utilize aim-assist cheats
- Cheat sophistications are black-box

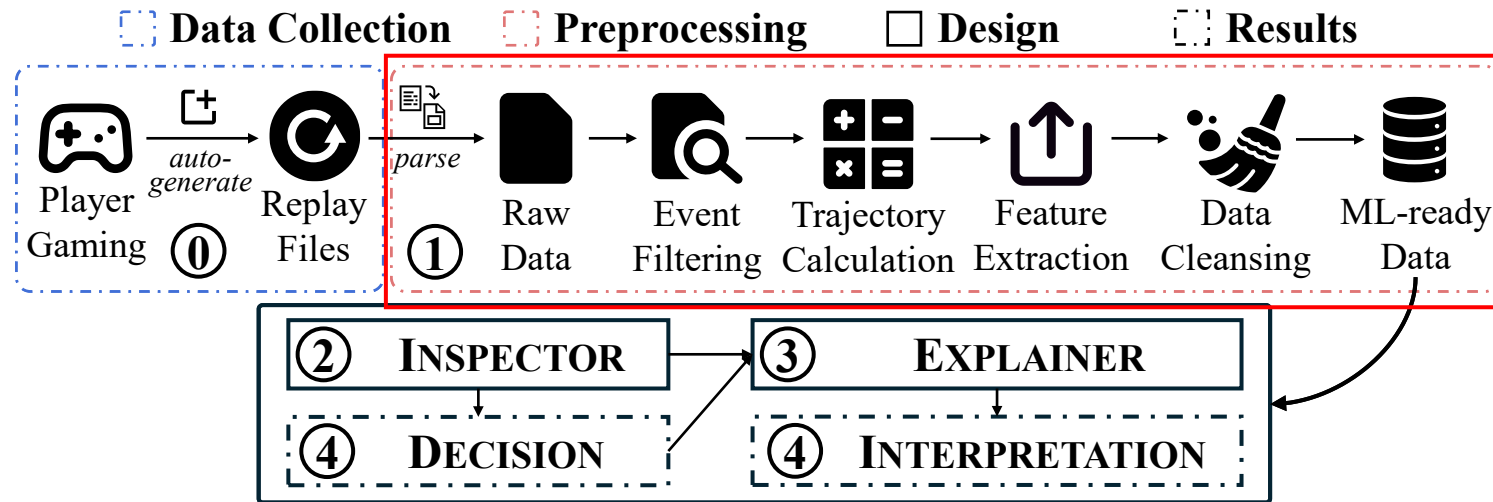
# Workflow



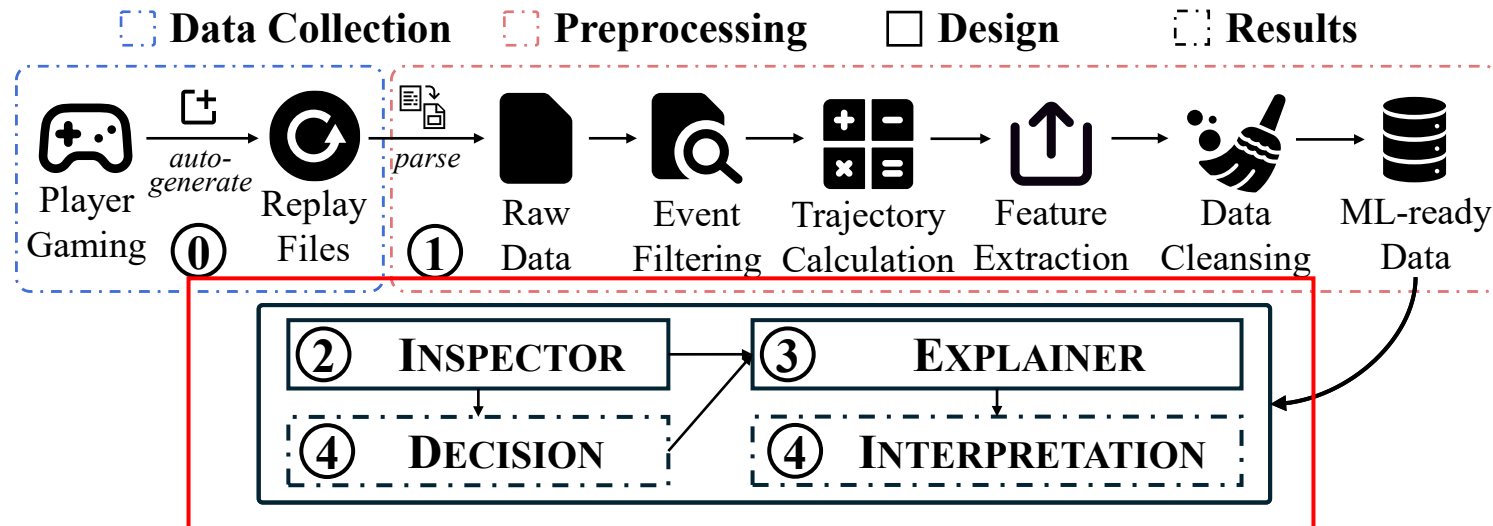
# Workflow



# Workflow



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# Dimensions of movement in FPS games

## Pitch and Yaw

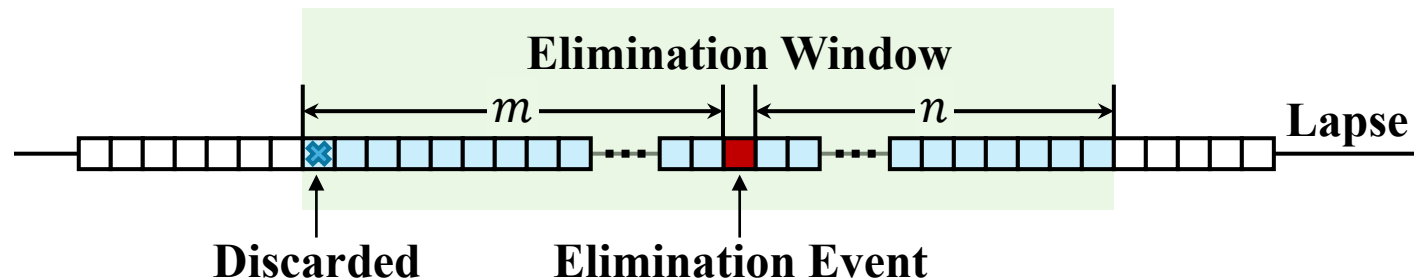
- **Pitch** is the vertical angle of the player's view.
- **Yaw** is the horizontal angle of the player's view.



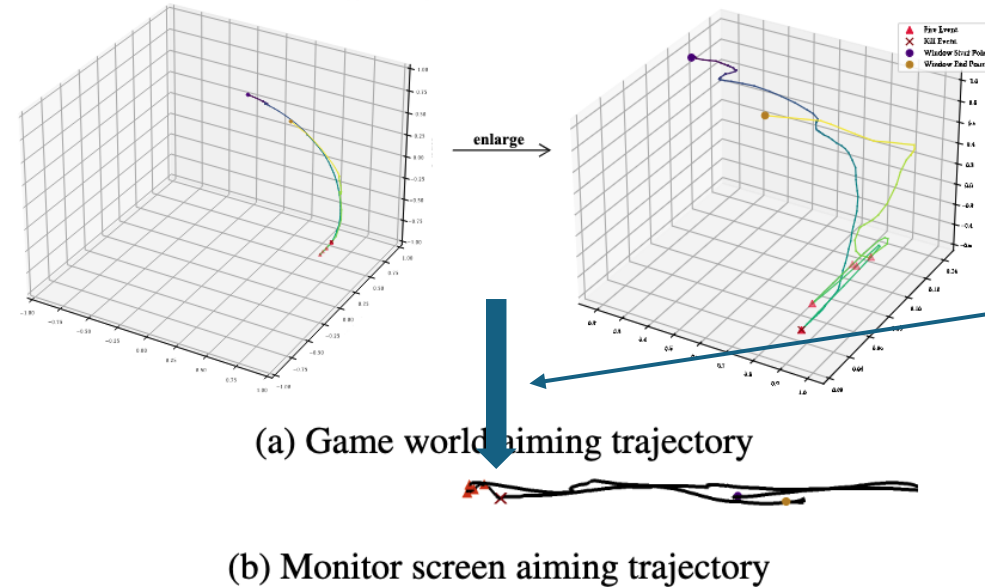
# Extraction scope – per elimination

## Elimination Window

- Created when a player triggers an elimination event (conduct a kill on an opponent).
- Size is 96.



# 3D aiming trajectory extraction and 2D mapping



## Algorithm 1 Convert Pitch and Yaw to 2D Coordinates

```

1: function PITCHYAWTOXY(pitch, yaw, width, height)
2:    $x \leftarrow \left( \frac{yaw+180}{360} \right) \times width$ 
3:    $y \leftarrow \left( \frac{pitch+90}{180} \right) \times height$ 
4:    $y \leftarrow height - y$ 
5:   return (x,y)
6: end function
  
```

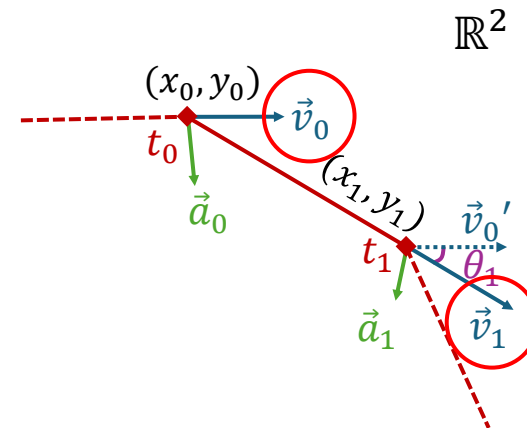
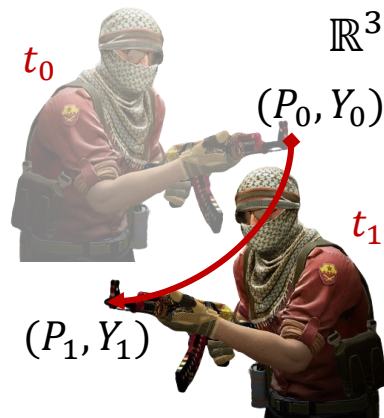
Figure 4: The example of 3D trajectory (a) and 2D screen trajectory mapping (b). The enlarged 3D trajectory (right) uses different measurement units for each coordinate axis for better illustration. 2D trajectory visualization is mapped onto a 1920×1080 resolution, and the coordinates of the elimination event are relocated to the center of the screen.

# Feature construction

## Velocity

$$\vec{v}_x = \frac{dx}{dt} \quad (1)$$

$$\vec{v}_y = \frac{dy}{dt} \quad (2)$$

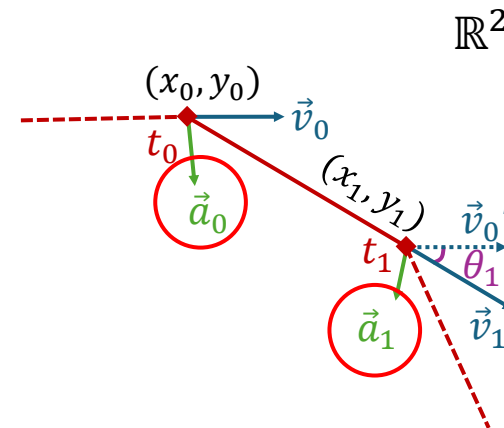
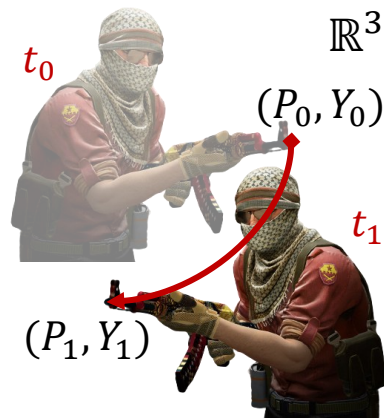


# Feature construction

## Acceleration

$$\vec{a}_x = \frac{d\vec{v}_x}{dt} = \frac{d^2x}{dt^2} \quad (3)$$

$$\vec{a}_y = \frac{d\vec{v}_y}{dt} = \frac{d^2y}{dt^2} \quad (4)$$

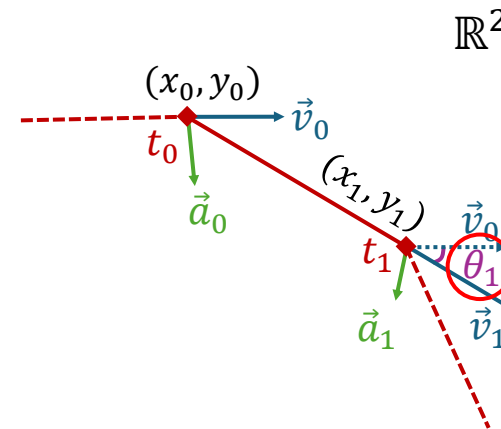
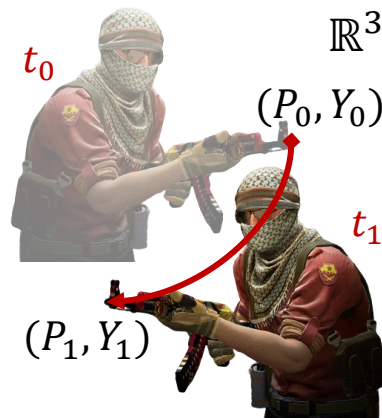


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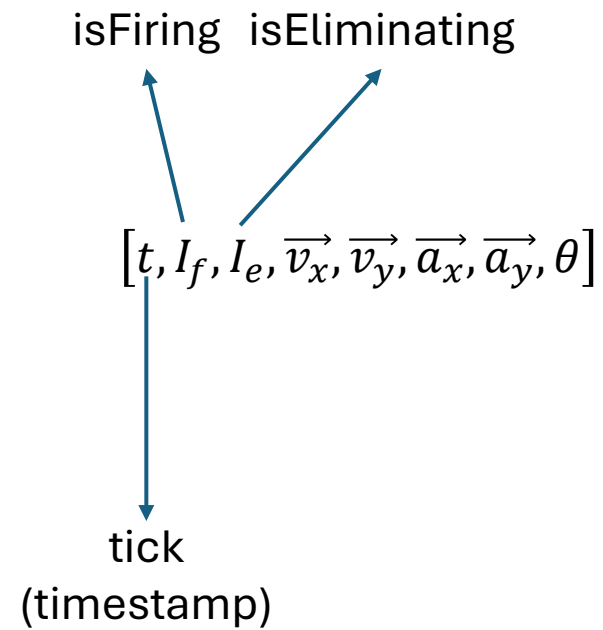
## Angular Change

$$\alpha = \arctan 2(\vec{v}_y, \vec{v}_x) \quad (5)$$

$$\theta = \frac{d\alpha}{dt} \quad (6)$$



# ML-ready data



# Framework

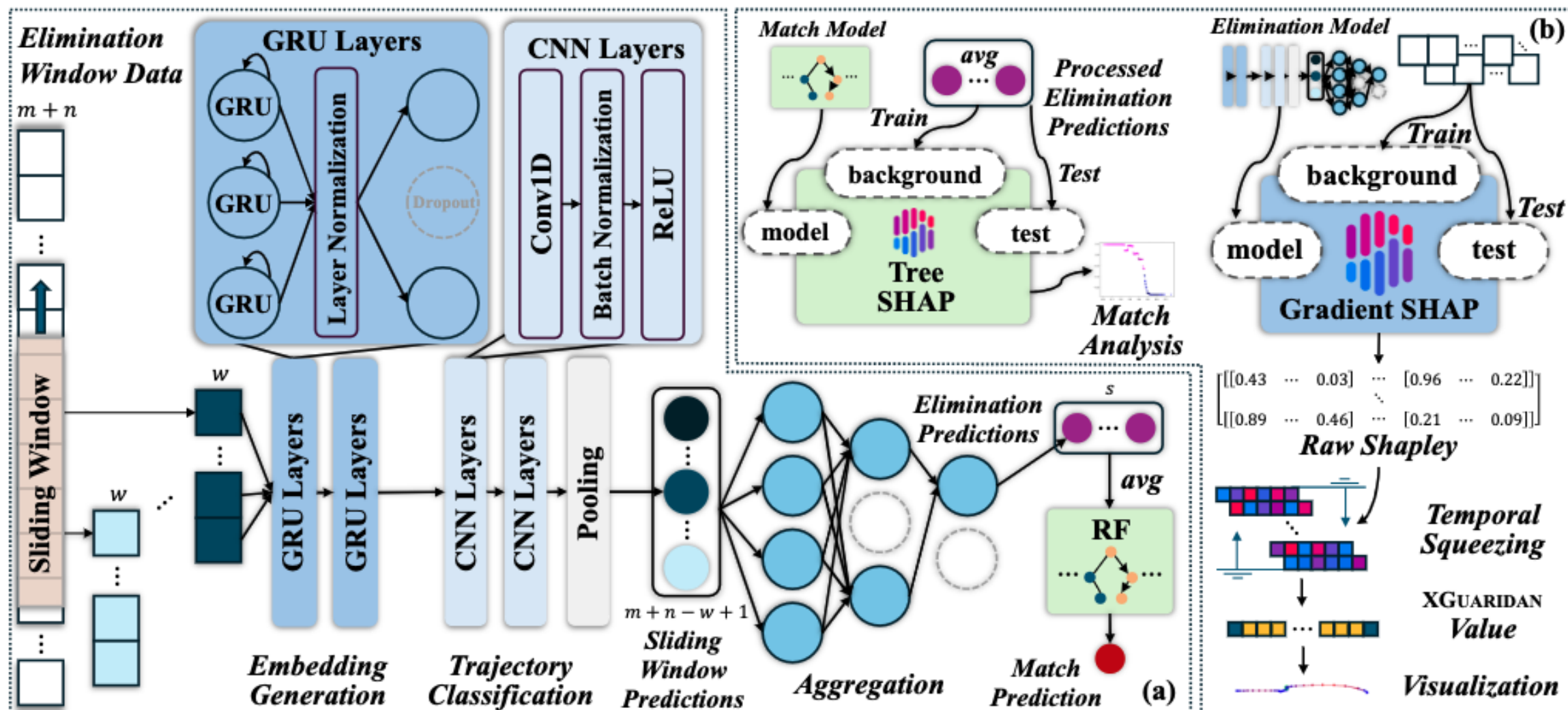
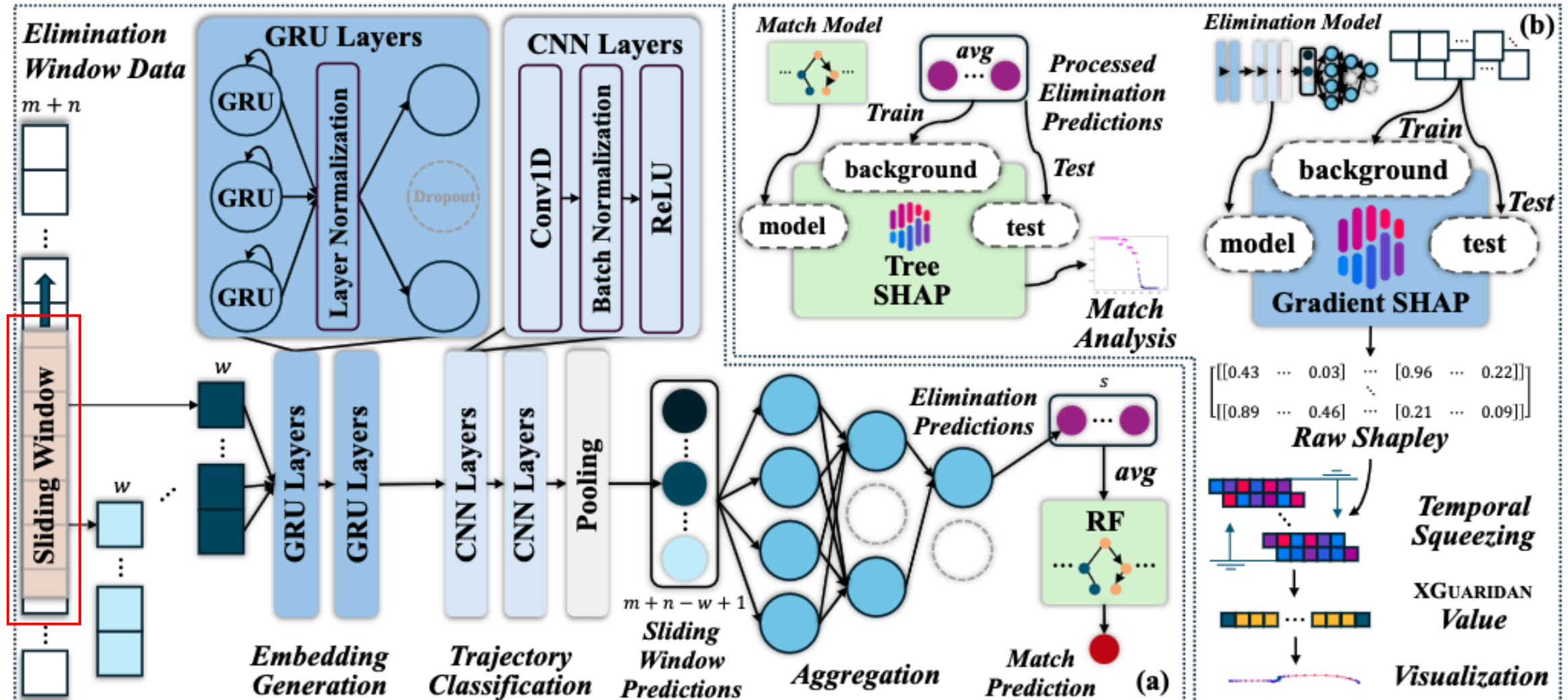


Figure 5: XGUARDIAN framework. The subfigures illustrate the architectures of (a)-INSPECTOR and (b)-EXPLAINER.



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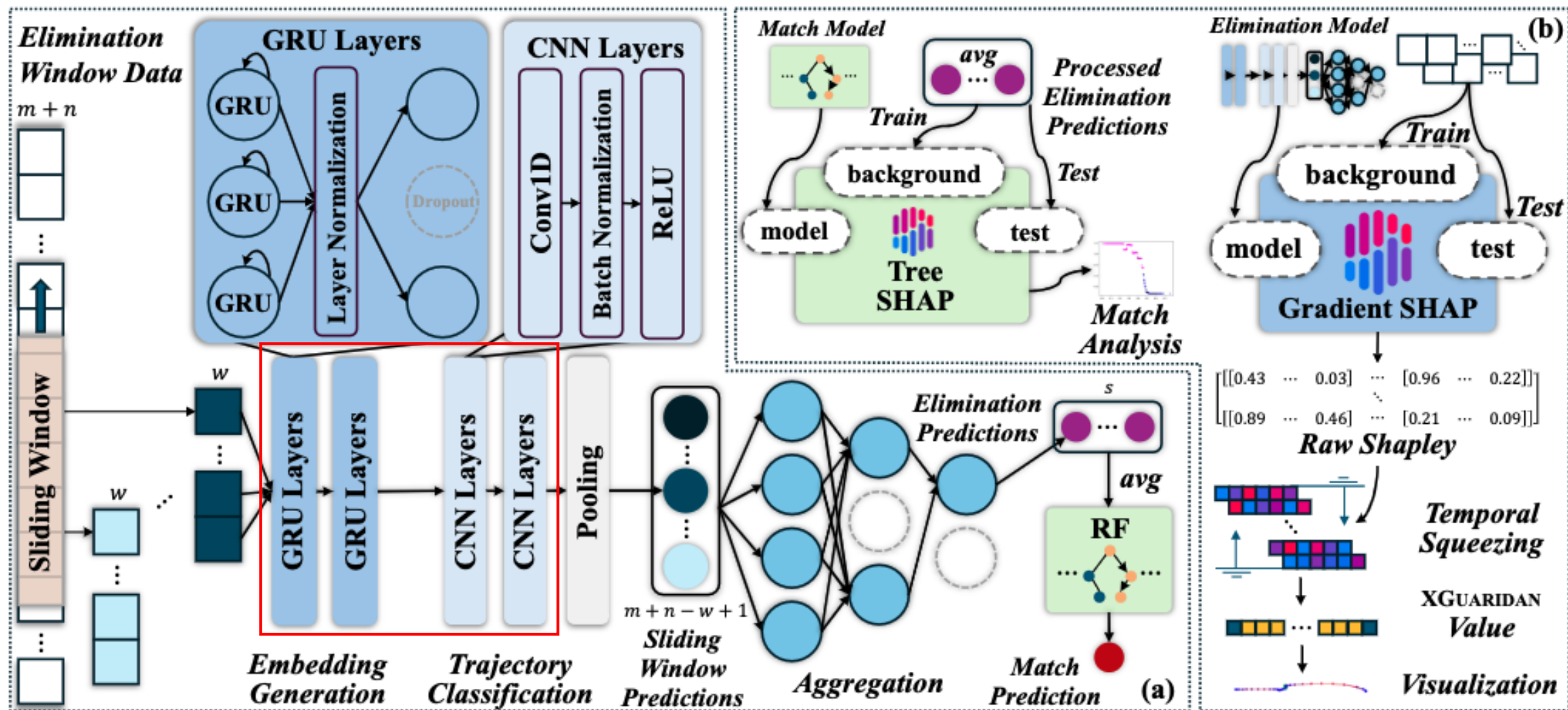


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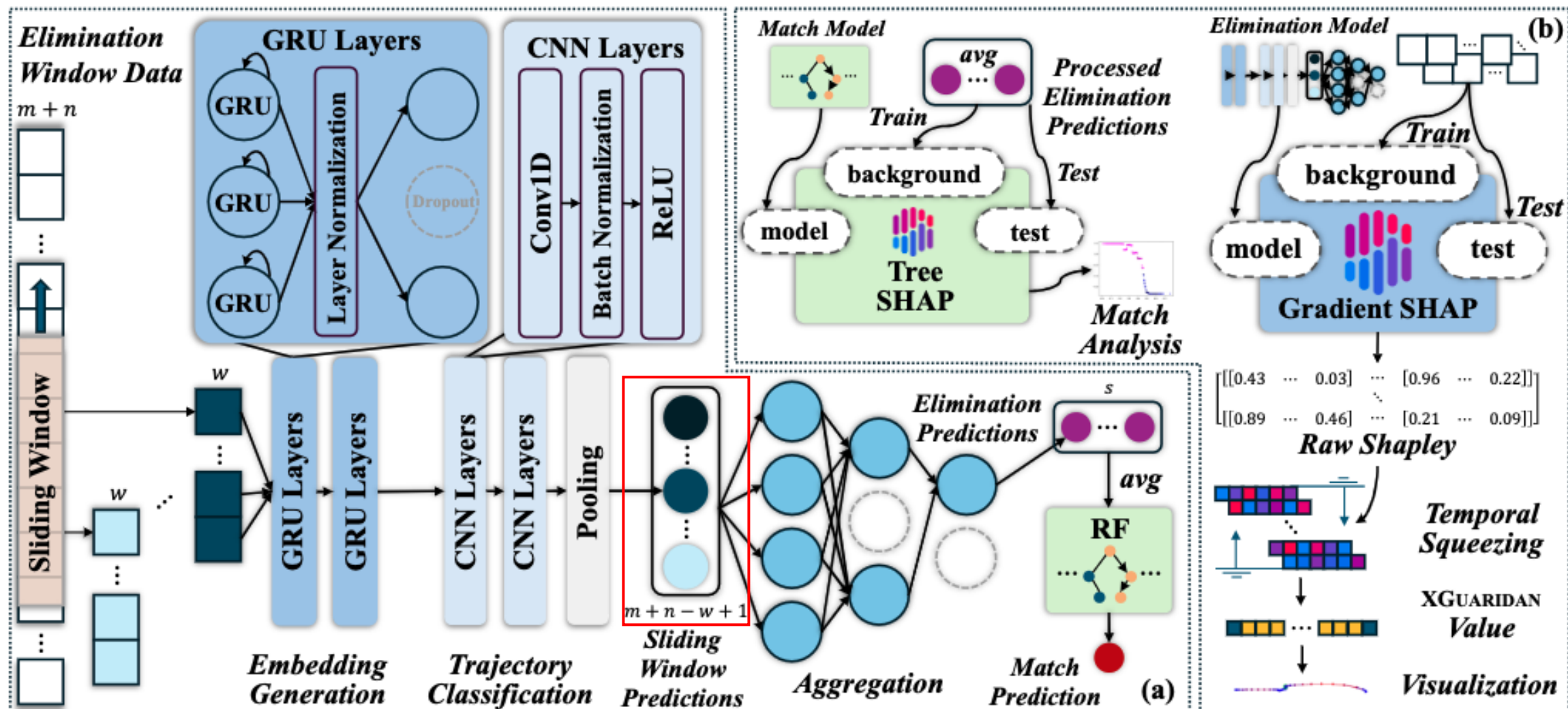


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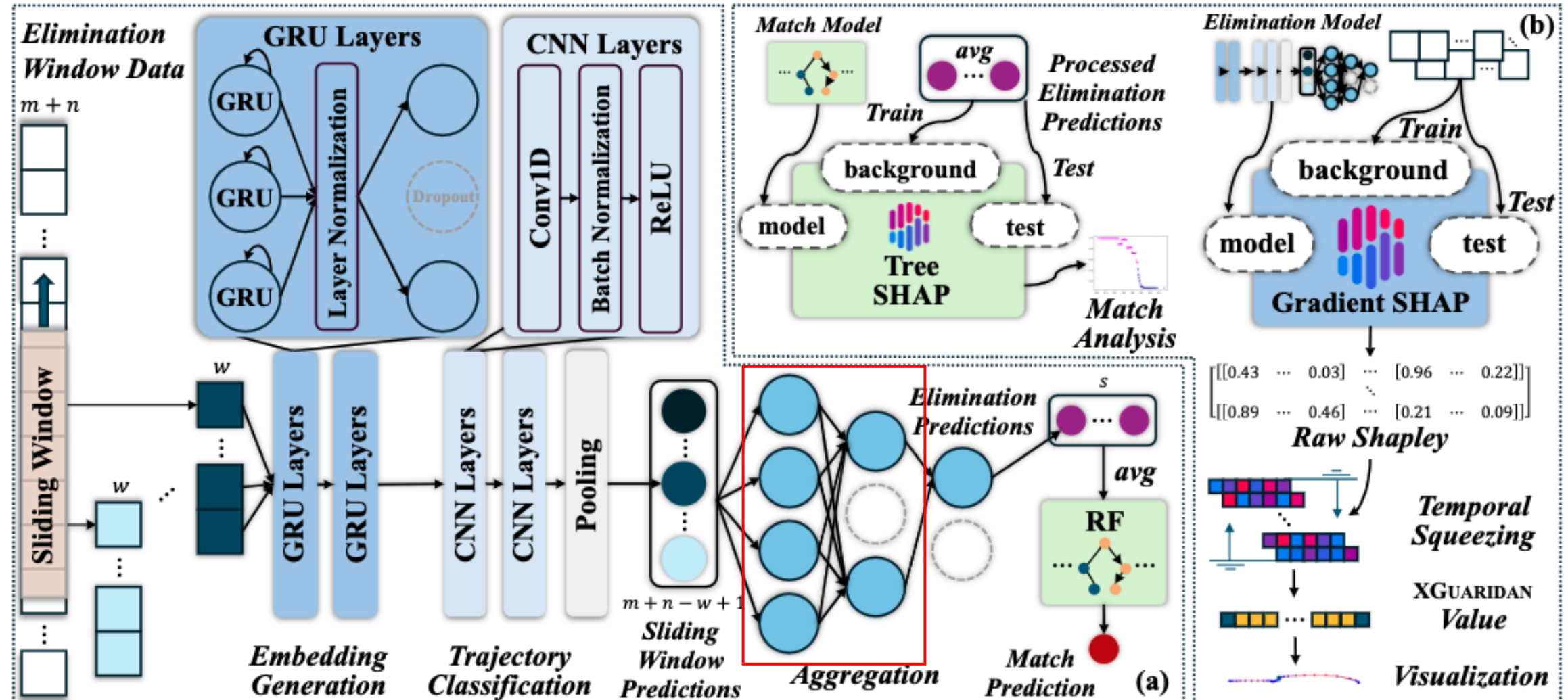
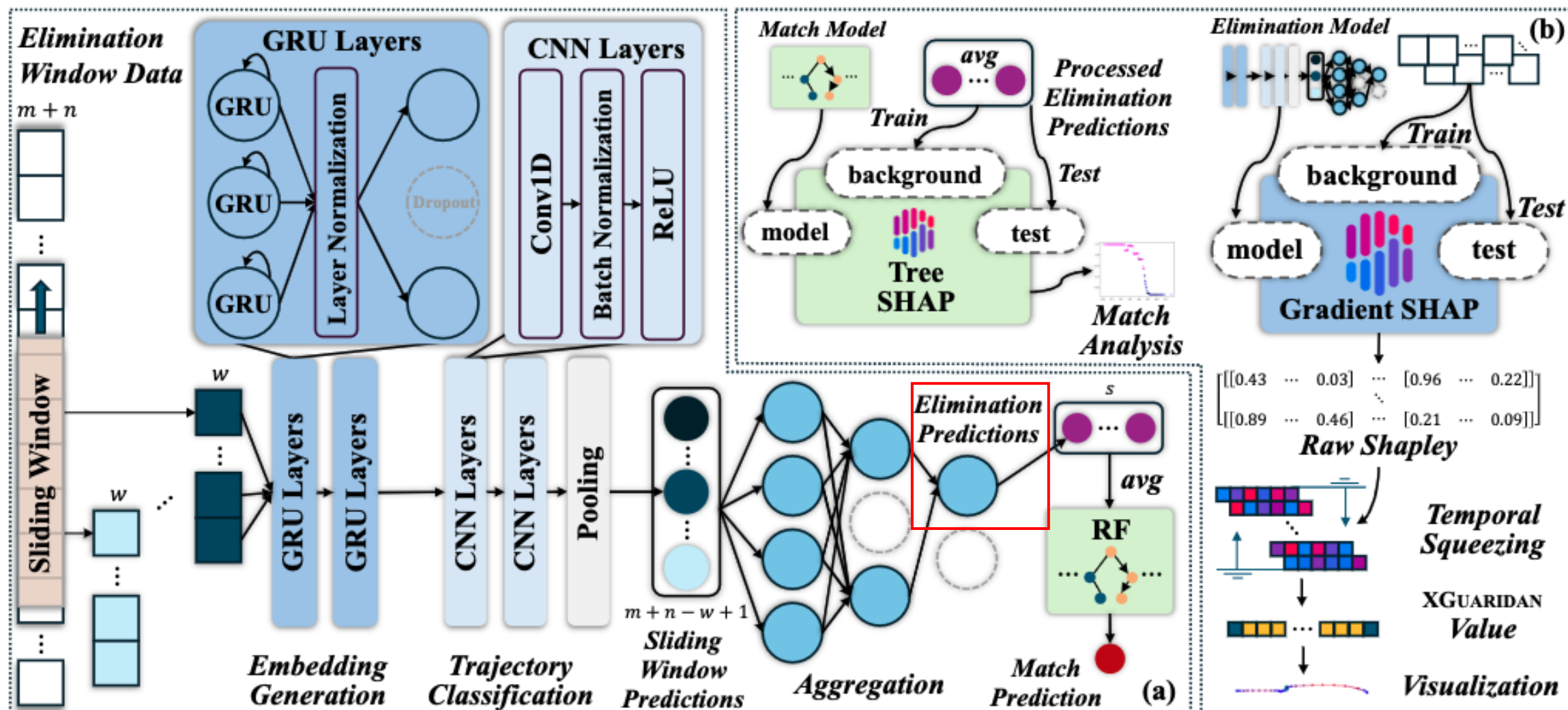


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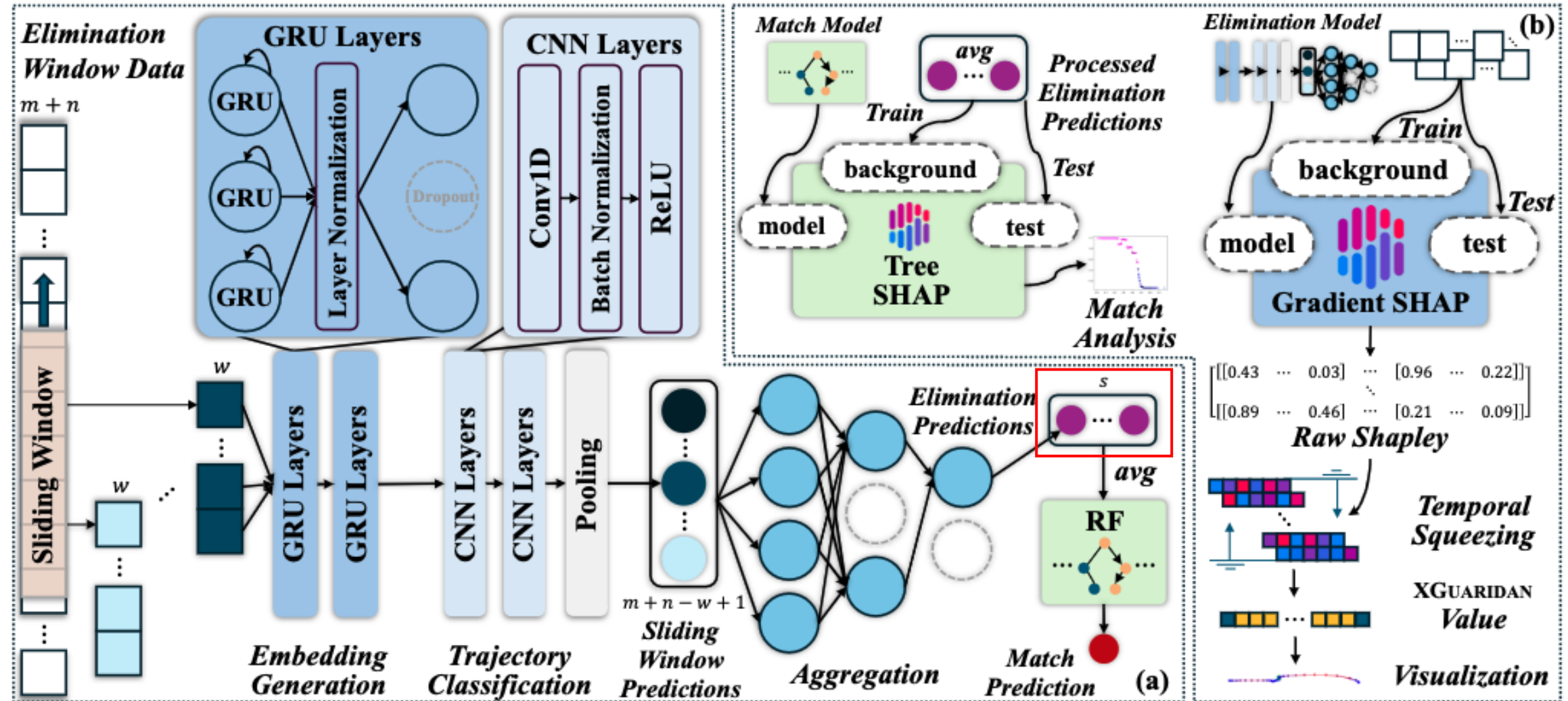
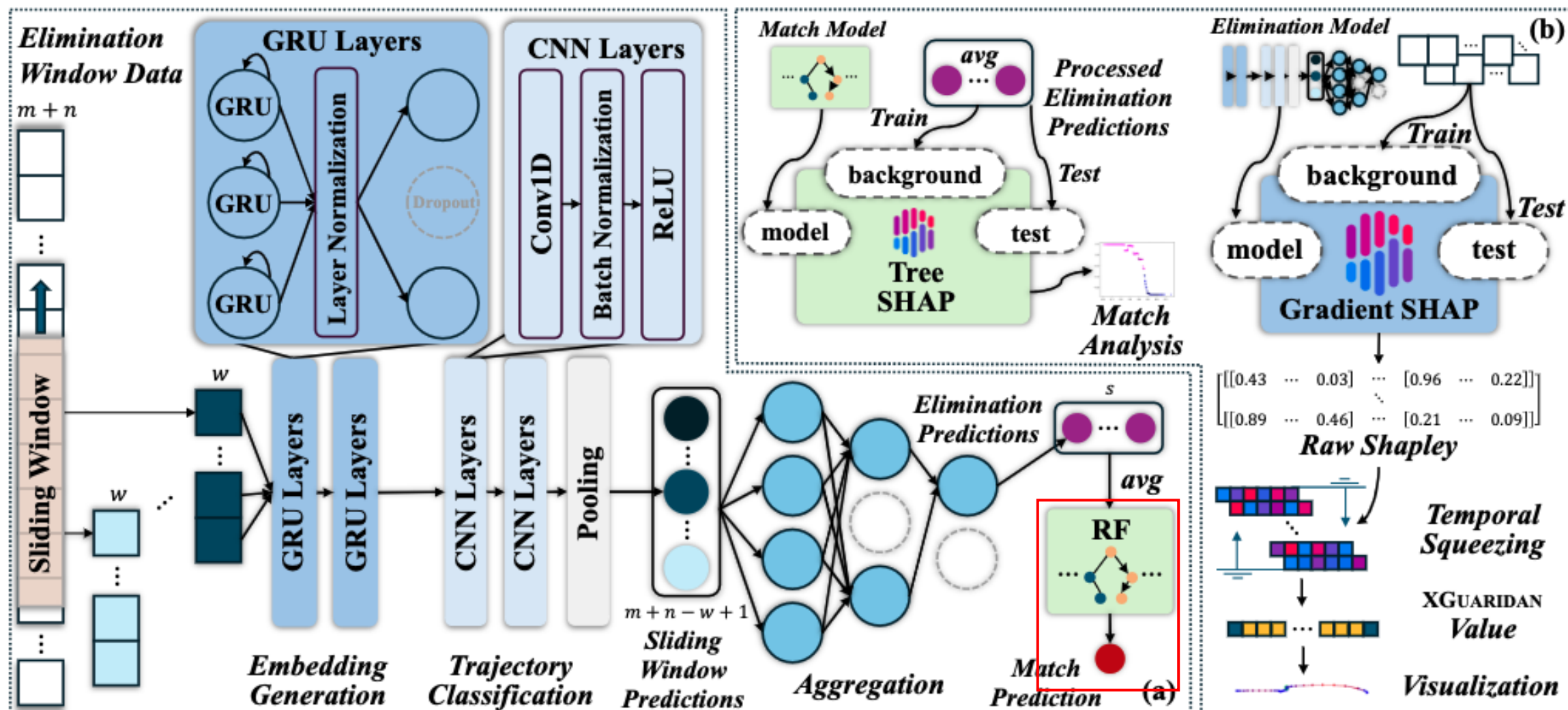


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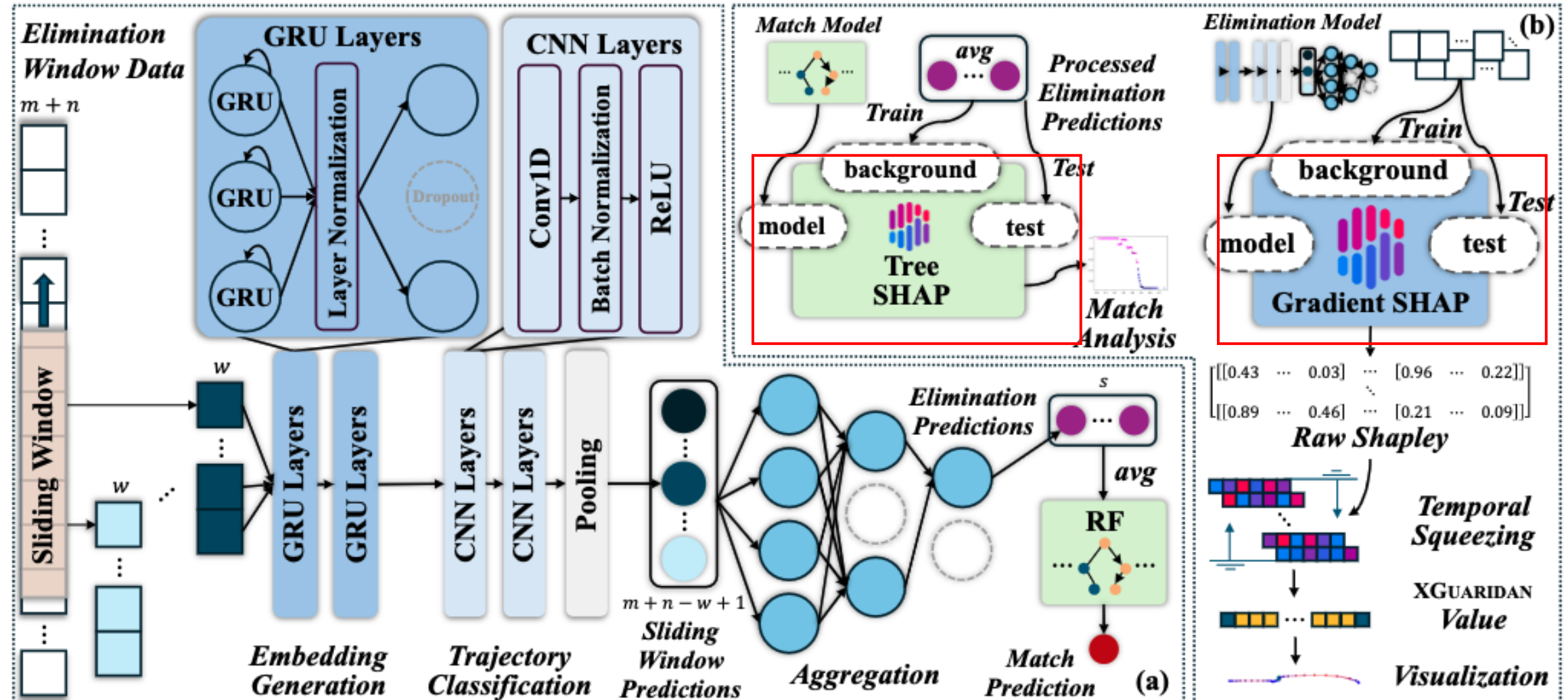


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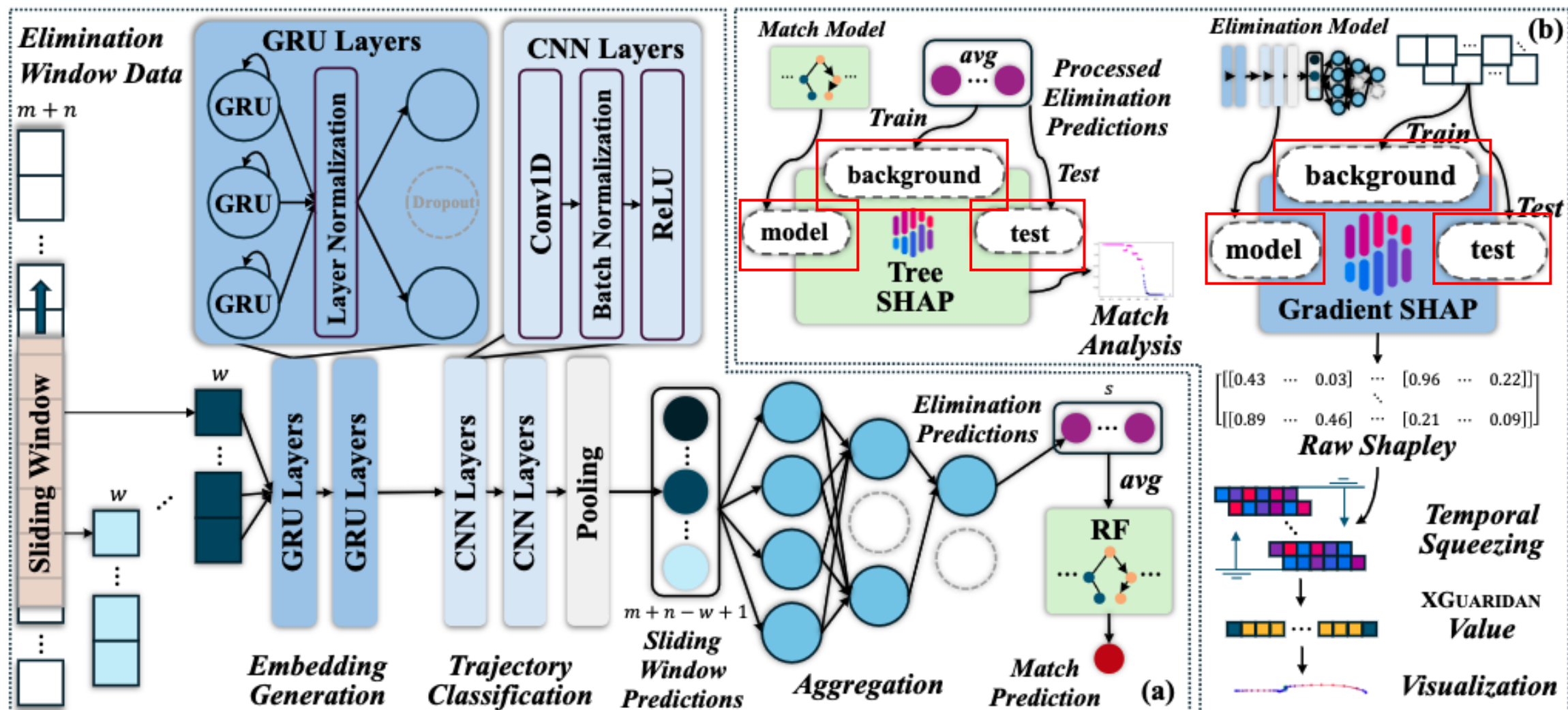
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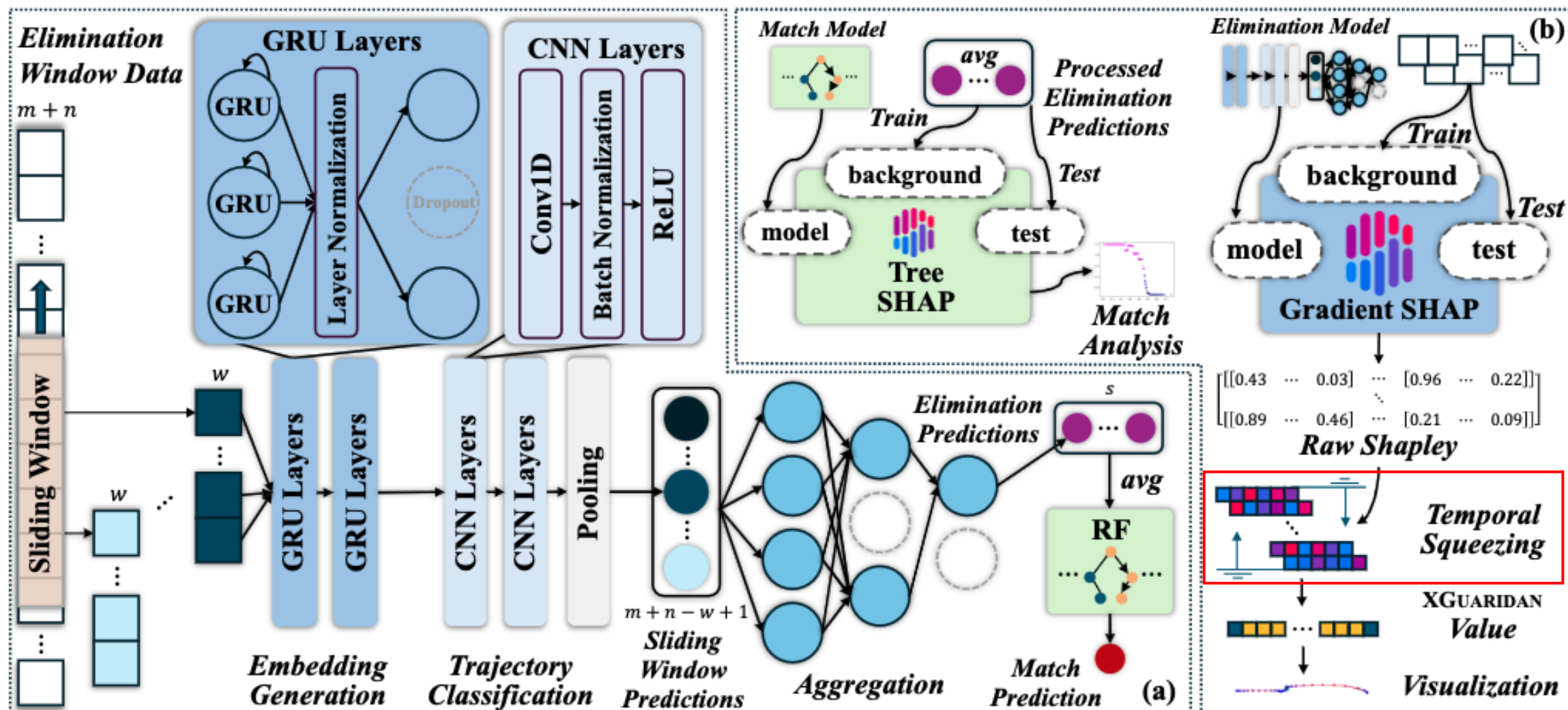
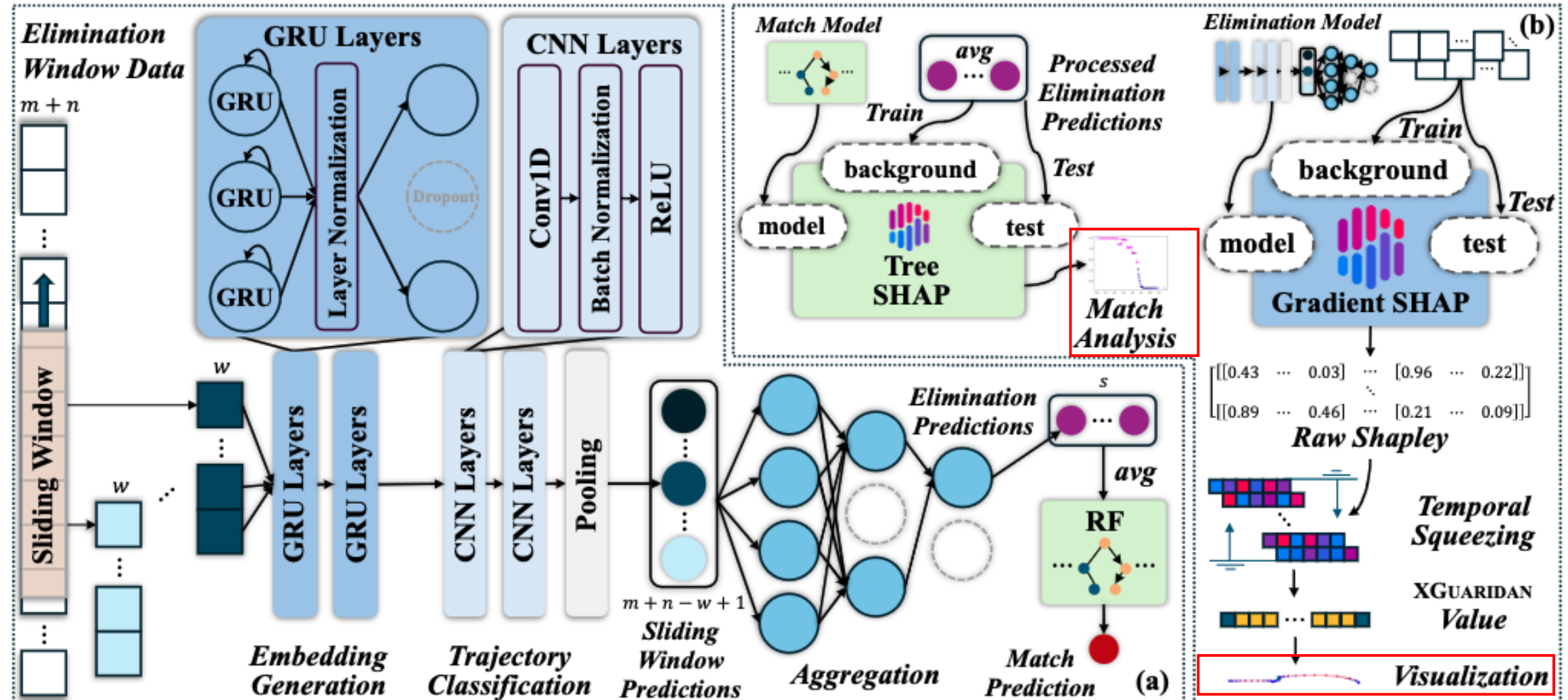


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# Dataset

Data is from *5EPlay*, a world-class CS2 platform.

Table 2: Dataset description.

| Data Split                          | A       | B         | C         |
|-------------------------------------|---------|-----------|-----------|
| <b>#Tick</b>                        | 992,064 | 1,014,624 | 1,062,528 |
| <b>#Total Elimination Window</b>    | 10,334  | 10,569    | 11,068    |
| <b>#Cheating Elimination Window</b> | 2,385   | 2,760     | 2,900     |
| <b>#Normal Elimination Window</b>   | 7,949   | 7,809     | 8,168     |
| <b>#Match</b>                       | 971     | 944       | 988       |
| <b>#Cheater</b>                     | 108     | 115       | 127       |
| <b>#Normal Player</b>               | 1,711   | 1,671     | 1,754     |
| <b>#Total Player</b>                | 1,819   | 1,786     | 1,881     |

# Detection Performance

Table 3: XGUARDIAN weighted average overall performance with different set combinations. **Blue**, **green**, and **red** in the combination rows indicate the use of the set as training, validation, and test set, respectively.

|                    |                            | Accuracy( $\uparrow$ ) | FPR( $\downarrow$ ) | Recall( $\uparrow$ ) | Precision( $\uparrow$ ) | F1-Score( $\uparrow$ ) |
|--------------------|----------------------------|------------------------|---------------------|----------------------|-------------------------|------------------------|
| Combination        | <b>A</b> <b>B</b> <b>C</b> | 0.889                  | 0.062               | 0.889                | 0.938                   | 0.907                  |
|                    | <b>A</b> <b>C</b> <b>B</b> | 0.876                  | 0.049               | 0.876                | 0.951                   | 0.901                  |
|                    | <b>B</b> <b>A</b> <b>C</b> | 0.882                  | 0.101               | 0.882                | 0.899                   | 0.890                  |
|                    | <b>B</b> <b>C</b> <b>A</b> | 0.907                  | 0.041               | 0.907                | 0.959                   | 0.924                  |
|                    | <b>C</b> <b>A</b> <b>B</b> | 0.872                  | 0.050               | 0.872                | 0.950                   | 0.898                  |
|                    | <b>C</b> <b>B</b> <b>A</b> | 0.879                  | 0.048               | 0.879                | 0.952                   | 0.904                  |
| mean ( $\mu$ )     |                            | 0.884                  | 0.059               | 0.884                | 0.942                   | 0.904                  |
| var ( $\sigma^2$ ) |                            | 1.32E-04               | 4.00E-04            | 1.32E-04             | 4.00E-04                | 1.08E-04               |
| std ( $\sigma$ )   |                            | 1.15E-02               | 2.00E-02            | 1.15E-02             | 2.00E-02                | 1.04E-02               |

# Detection Performance

Table 3: XGUARDIAN weighted average overall performance with different set combinations. **Blue**, **green**, and **red** in the combination rows indicate the use of the set as training, validation, and test set, respectively.

|                                    |            | Accuracy( $\uparrow$ ) | FPR( $\downarrow$ ) | Recall( $\uparrow$ ) | Precision( $\uparrow$ ) | F1-Score( $\uparrow$ ) |
|------------------------------------|------------|------------------------|---------------------|----------------------|-------------------------|------------------------|
| Combination                        | <b>ABC</b> | 0.889                  | 0.062               | 0.889                | 0.938                   | 0.907                  |
|                                    | <b>ACB</b> | 0.876                  | 0.049               | 0.876                | 0.951                   | 0.901                  |
|                                    | <b>BAC</b> | 0.882                  | 0.101               | 0.882                | 0.899                   | 0.890                  |
|                                    | <b>BCA</b> | 0.907                  | 0.041               | 0.907                | 0.959                   | 0.924                  |
|                                    | <b>CAB</b> | 0.872                  | 0.050               | 0.872                | 0.950                   | 0.898                  |
|                                    | <b>CBA</b> | 0.879                  | 0.048               | 0.879                | 0.952                   | 0.904                  |
| <b>mean (<math>\mu</math>)</b>     |            | 0.884                  | 0.059               | 0.884                | 0.942                   | 0.904                  |
| <b>var (<math>\sigma^2</math>)</b> |            | 1.32E-04               | 4.00E-04            | 1.32E-04             | 4.00E-04                | 1.08E-04               |
| <b>std (<math>\sigma</math>)</b>   |            | 1.15E-02               | 2.00E-02            | 1.15E-02             | 2.00E-02                | 1.04E-02               |



# Detection Performance

Table 3: XGUARDIAN weighted average overall performance with different set combinations. **Blue**, **green**, and **red** in the combination rows indicate the use of the set as training, validation, and test set, respectively.

|                    |            | Accuracy( $\uparrow$ ) | FPR( $\downarrow$ ) | Recall( $\uparrow$ ) | Precision( $\uparrow$ ) | F1-Score( $\uparrow$ ) |
|--------------------|------------|------------------------|---------------------|----------------------|-------------------------|------------------------|
| Combination        | <b>ABC</b> | 0.889                  | 0.062               | 0.889                | 0.938                   | 0.907                  |
|                    | <b>ACB</b> | 0.876                  | 0.049               | 0.876                | 0.951                   | 0.901                  |
|                    | <b>BAC</b> | 0.882                  | 0.101               | 0.882                | 0.899                   | 0.890                  |
|                    | <b>BCA</b> | 0.907                  | 0.041               | 0.907                | 0.959                   | 0.924                  |
|                    | <b>CAB</b> | 0.872                  | 0.050               | 0.872                | 0.950                   | 0.898                  |
|                    | <b>CBA</b> | 0.879                  | 0.048               | 0.879                | 0.952                   | 0.904                  |
| mean ( $\mu$ )     |            | 0.884                  | 0.059               | 0.884                | 0.942                   | 0.904                  |
| var ( $\sigma^2$ ) |            | 1.32E-04               | 4.00E-04            | 1.32E-04             | 4.00E-04                | 1.08E-04               |
| std ( $\sigma$ )   |            | 1.15E-02               | 2.00E-02            | 1.15E-02             | 2.00E-02                | 1.04E-02               |



# Feature ablation study

$$[t, I_f, I_e, \vec{v}_x, \vec{v}_y, \vec{a}_x, \vec{a}_y, \theta]$$

Table 4: Feature ablation study weighted average results.

| Feature                |                        |          | Metrics  |       |        |           |          |
|------------------------|------------------------|----------|----------|-------|--------|-----------|----------|
| $\vec{a}_x, \vec{a}_y$ | $\vec{v}_x, \vec{v}_y$ | $\theta$ | Accuracy | FPR   | Recall | Precision | F1-Score |
| ✓                      | ✗                      | ✗        | 0.838    | 0.059 | 0.838  | 0.941     | 0.873    |
| ✗                      | ✓                      | ✗        | 0.884    | 0.060 | 0.884  | 0.940     | 0.904    |
| ✗                      | ✗                      | ✓        | 0.877    | 0.057 | 0.877  | 0.943     | 0.900    |
| ✓                      | ✓                      | ✗        | 0.886    | 0.062 | 0.886  | 0.938     | 0.905    |
| ✓                      | ✗                      | ✓        | 0.880    | 0.058 | 0.880  | 0.942     | 0.902    |
| ✗                      | ✓                      | ✓        | 0.901    | 0.063 | 0.901  | 0.937     | 0.915    |
| ✓                      | ✓                      | ✓        | 0.890    | 0.060 | 0.890  | 0.940     | 0.908    |

Table 5: XGUARDIAN input feature *tick* indexing comparative analysis weighted average results. ‘wo/’ and ‘w/’ denote without and with. ‘Reset index’ indicates re-allocating each elimination window’s tick index from 0 to  $m + n - 1$ .

| Tick Indexing                   | Accuracy     | FPR          | Recall       | Precision    | F1-Score     |
|---------------------------------|--------------|--------------|--------------|--------------|--------------|
| wo/ tick index                  | 0.801        | 0.078        | 0.801        | 0.922        | 0.846        |
| w/ reset index                  | 0.784        | 0.073        | 0.784        | 0.927        | 0.834        |
| <b>XGUARDIAN (w/ raw index)</b> | <b>0.889</b> | <b>0.062</b> | <b>0.889</b> | <b>0.938</b> | <b>0.907</b> |

# Inspector model and parameters

- Different elimination window sizes
- Different sliding window sizes

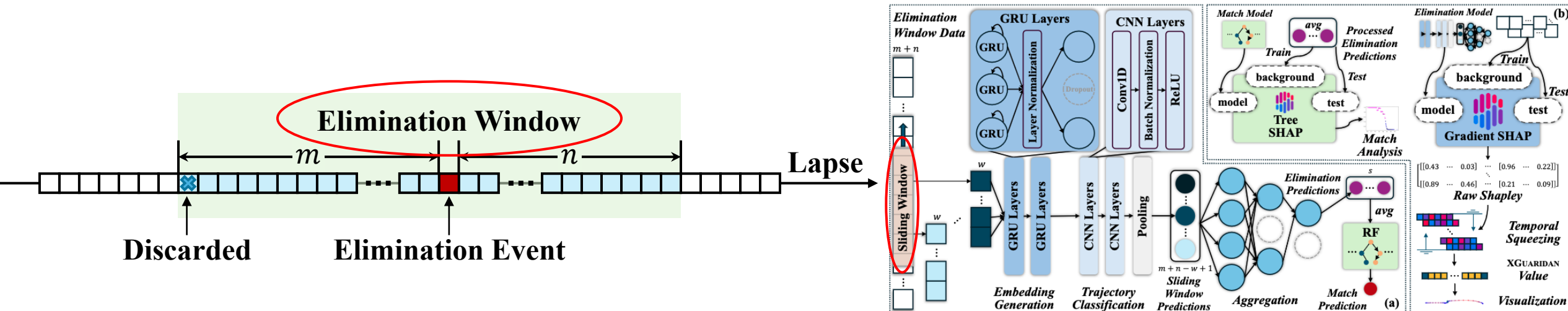


Figure 5: XGUARDIAN framework. The subfigures illustrate the architectures of (a)-INSPECTOR and (b)-EXPLAINER.

# Inspector model and parameters

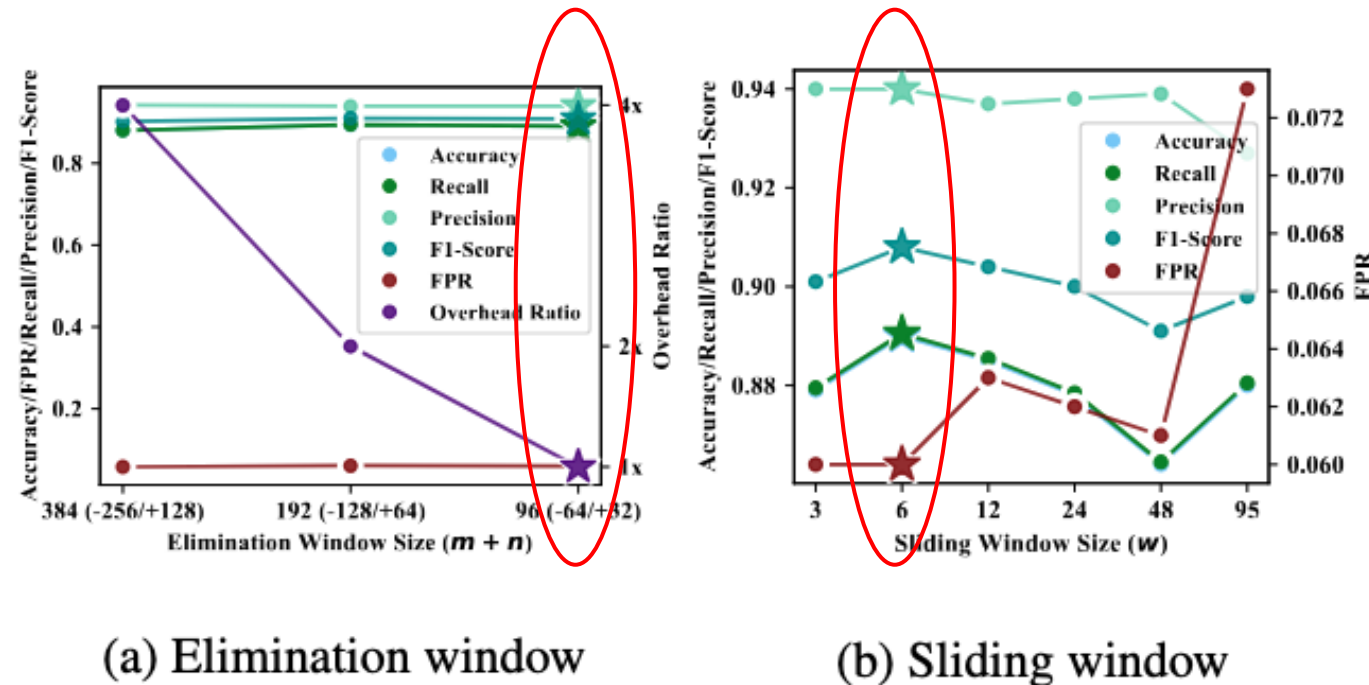


Figure 6: INSPECTOR sliding window and elimination window size comparative analysis weighted average results. The star (★) indicates the parameters used by XGUARDIAN.

# Inspector model and parameters

- Different models in Embedding Generation
- Different models in Trajectory Classification

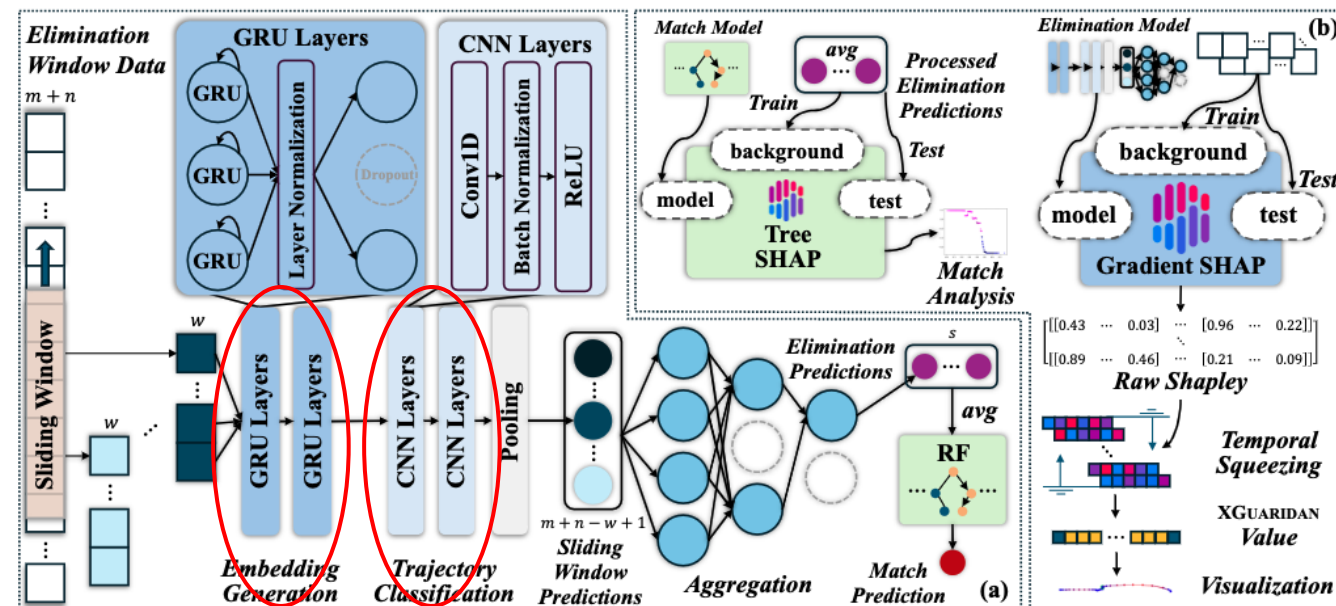
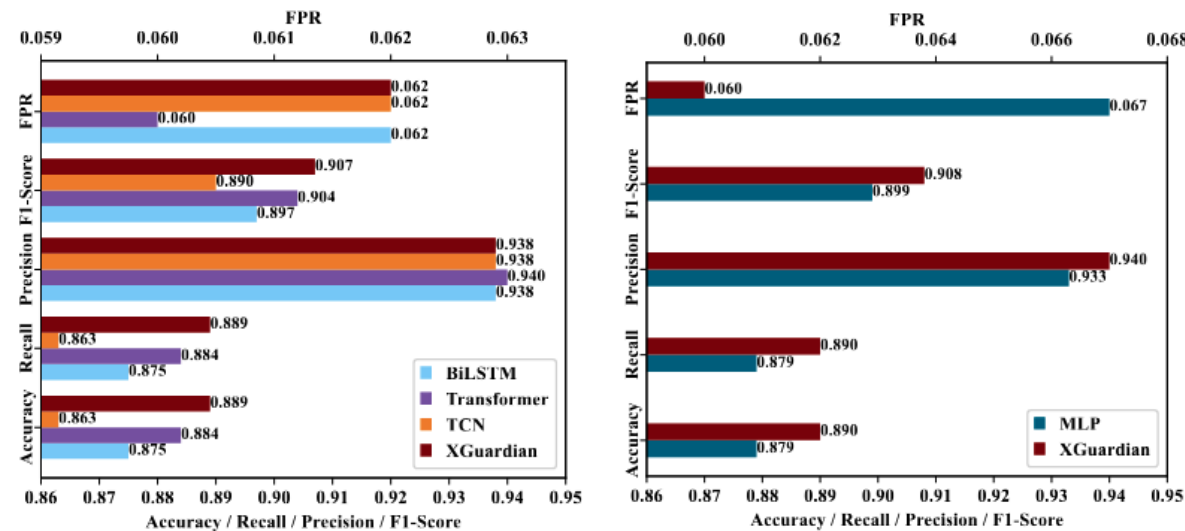


Figure 5: XGUARDIAN framework. The subfigures illustrate the architectures of (a)-INSPECTOR and (b)-EXPLAINER.

# Inspector model and parameters



(a) Embedding generation stage (b) Trajectory classification stage

Figure 7: INSPECTOR embedding generation and trajectory classification stages model selection weighted average results across SOTA models [30–33].

# Inspector model and parameters

- Different input styles in Elimination Predictions

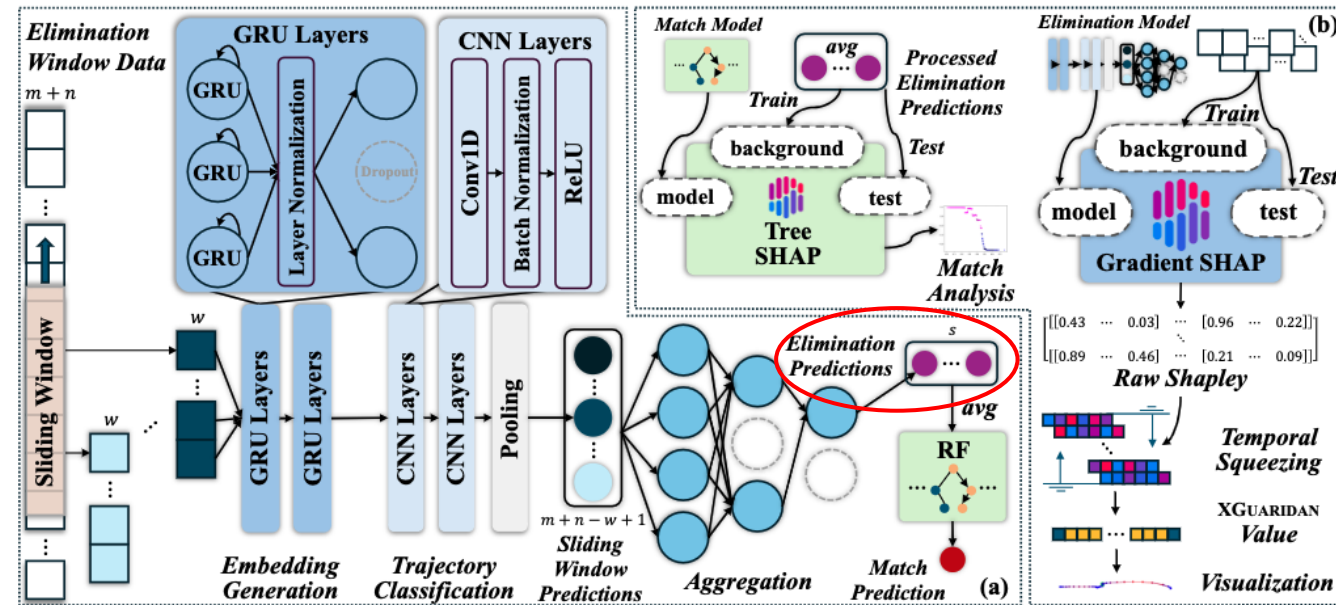


Figure 5: XGUARDIAN framework. The subfigures illustrate the architectures of (a)-INSPECTOR and (b)-EXPLAINER.

# Inspector model and parameters

Table 6: INSPECTOR elimination predictions stage embedding style comparison. ‘Binary’ denotes embedding binary classified predictions as match prediction’s input. ‘Original’ denotes embedding raw predictions as match prediction input. XGUARDIAN’s configuration is bolded.

| Input Style                 | Accuracy     | FPR          | Recall       | Precision    | F1-Score     |
|-----------------------------|--------------|--------------|--------------|--------------|--------------|
| Binary                      | 0.881        | 0.061        | 0.881        | 0.939        | 0.902        |
| Binary+Original             | 0.881        | 0.061        | 0.881        | 0.939        | 0.902        |
| <b>XGUARDIAN (Original)</b> | <b>0.889</b> | <b>0.062</b> | <b>0.889</b> | <b>0.938</b> | <b>0.907</b> |



# Comparative Study

Table 7: XGUARDIAN comparative study to prior works using different data sources. The optimal results are bolded.

| Dataset          | Method           | Accuracy     | FPR          | Recall       | Precision    | F1-Score     |
|------------------|------------------|--------------|--------------|--------------|--------------|--------------|
| <b>XGUARDIAN</b> | TH_ACCA [7]      | 0.738        | 0.455        | 0.738        | 0.545        | 0.627        |
|                  | TH_HITACC [10]   | 0.644        | 0.086        | 0.644        | 0.914        | 0.711        |
|                  | <b>XGUARDIAN</b> | <b>0.889</b> | <b>0.062</b> | <b>0.889</b> | <b>0.938</b> | <b>0.907</b> |
| <b>HAWK</b> [4]  | REVPOV [4]       | 0.616        | 0.151        | 0.616        | 0.849        | 0.694        |
|                  | REVSTATS [4]     | 0.814        | 0.109        | 0.814        | 0.891        | 0.842        |
|                  | ExSPC [4]        | 0.793        | 0.114        | 0.793        | 0.886        | 0.827        |
|                  | HAWK [4]         | 0.716        | 0.112        | 0.716        | 0.888        | 0.772        |
|                  | <b>XGUARDIAN</b> | <b>0.841</b> | <b>0.056</b> | <b>0.841</b> | <b>0.944</b> | <b>0.878</b> |

# Comparative Study

Table 7: XGUARDIAN comparative study to prior works using different data sources. The optimal results are bolded.

| Dataset   | Method         | Accuracy     | FPR          | Recall       | Precision    | F1-Score     |
|-----------|----------------|--------------|--------------|--------------|--------------|--------------|
| XGUARDIAN | TH_ACCA [7]    | 0.738        | 0.455        | 0.738        | 0.545        | 0.627        |
|           | TH_HITACC [10] | 0.644        | 0.086        | 0.644        | 0.914        | 0.711        |
|           | XGUARDIAN      | <b>0.889</b> | <b>0.062</b> | <b>0.889</b> | <b>0.938</b> | <b>0.907</b> |
| HAWK [4]  | REVPOV [4]     | 0.616        | 0.151        | 0.616        | 0.849        | 0.694        |
|           | REVSTATS [4]   | 0.814        | 0.109        | 0.814        | 0.891        | 0.842        |
|           | ExSPC [4]      | 0.793        | 0.114        | 0.793        | 0.886        | 0.827        |
|           | HAWK [4]       | 0.716        | 0.112        | 0.716        | 0.888        | 0.772        |
|           | XGUARDIAN      | <b>0.841</b> | <b>0.056</b> | <b>0.841</b> | <b>0.944</b> | <b>0.878</b> |

# Generalizability Study

Table 8: XGUARDIAN’s generalizability evaluation results on different sub-genre games, platforms, and engines. The descriptions of the other two datasets are shown in [Appendix F](#).

| Dataset                    | HAWK [4]         | FARLIGHT84        | XGUARDIAN        |
|----------------------------|------------------|-------------------|------------------|
| <b>Game (Release Year)</b> | CS:GO (2012)     | Farlight84 (2023) | CS2 (2023)       |
| <b>Sub-genre</b>           | Tactical Shooter | Battle Royal      | Tactical Shooter |
| <b>Engine</b>              | Source           | Unreal Engine 4   | Source 2         |
| <b>Platform</b>            | PC               | Mobile            | PC               |
| <b>Accuracy</b>            | 0.841            | 0.816             | 0.889            |
| <b>FPR</b>                 | 0.056            | 0.144             | 0.062            |
| <b>Recall</b>              | 0.841            | 0.816             | 0.889            |
| <b>Precision</b>           | 0.944            | 0.856             | 0.938            |
| <b>F1-Score</b>            | 0.878            | 0.829             | 0.907            |

# Explainability Case Study

<https://xguardian-anti-cheat.github.io/>

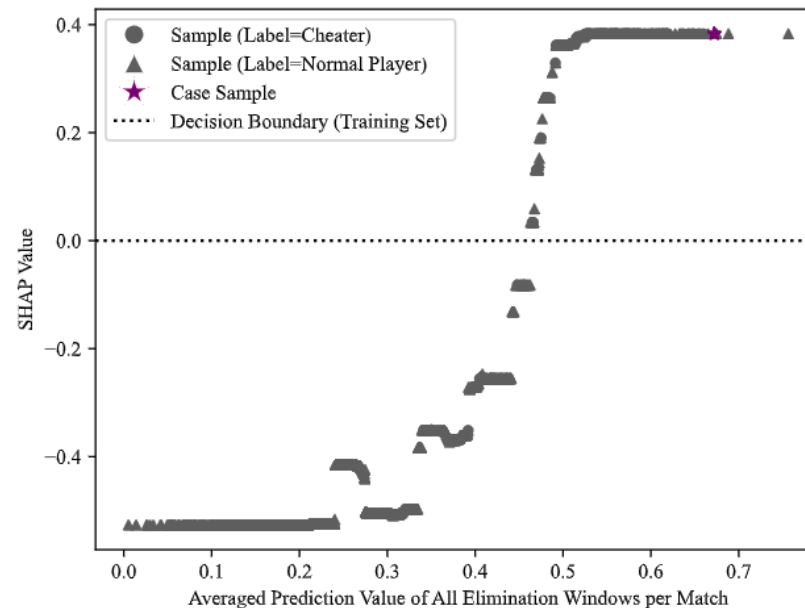


Figure 10: Explainable match-scope visualizations of an identified aim-assist cheater.

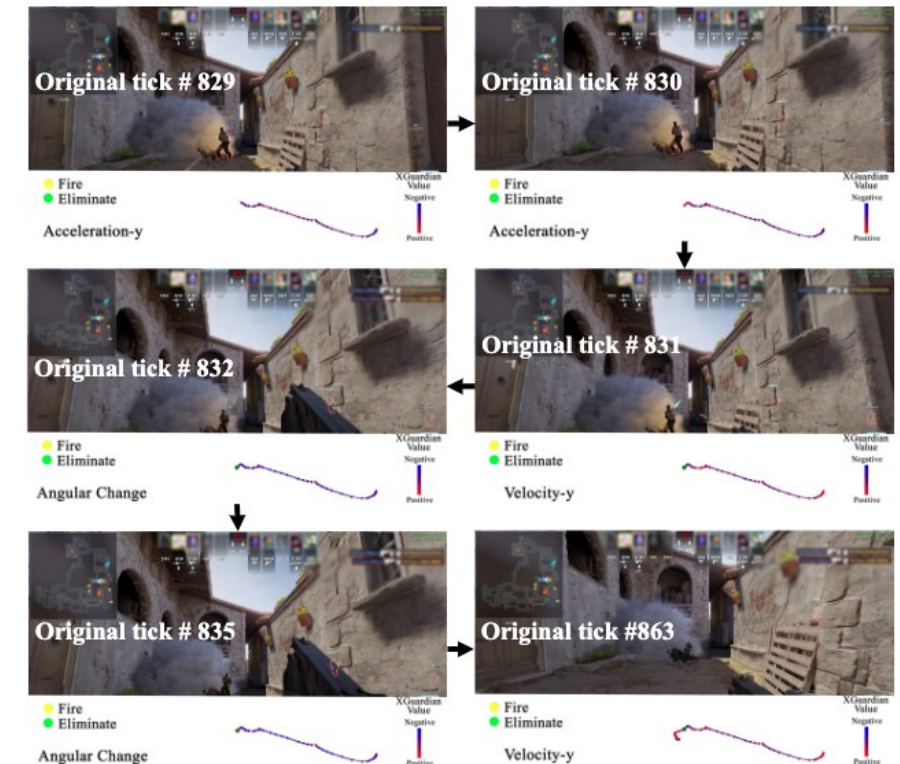


Figure 9: Explainable elimination-scope visualizations of an identified aim-assist cheater.

# Robustness Under Adversarial Attack

- **18/22** - Detected / All adversarial samples
  - achieves **0.818** *recall* and **0.900** *F1-Score*
- *Two-proportion Z-test* and *Fisher's exact test* indicate that the observed decrease in recall under adversarial attack is statistically **insignificant** ( $p > 0.05$ ).

# Overhead

- Only takes less than **9.98** seconds of prediction time per match with CPU-only
- A remarkable improvement of **4,104.31%** compared to the SOTA server-side solution HAWK, which requires over *400* seconds

Table 9: XGUARDIAN's overheads tested with the entire dataset. The numbers are retrieved by the Python libraries `time` [39] and `resource` [40].

| Overheads                   | Preprocessing | Prediction | Training |
|-----------------------------|---------------|------------|----------|
| User time (sec)             | 52.4          | 28966.87   | 28960.42 |
| System time (sec)           | 30.21         | 2905.86    | 2905.14  |
| Elapsed time (sec)          | 48.03         | 3.95       | 11423.11 |
| Max. resident set size (GB) | 3.226         | 8.596      | 8.596    |

# EXPLAINER’s Potential Bias Investigation

Table 11: User study results of the EXPLAINER’s potential bias questionnaire.

| Reviewer | Decision <sup>1</sup> |    |    |    |    |    | Confidence <sup>2</sup> |    |    |    |    |    | Time Consumption |       |       |       |       |       |
|----------|-----------------------|----|----|----|----|----|-------------------------|----|----|----|----|----|------------------|-------|-------|-------|-------|-------|
|          | 1A                    | 1B | 2A | 2B | 3A | 3B | 1A                      | 1B | 2A | 2B | 3A | 3B | 1A               | 1B    | 2A    | 2B    | 3A    | 3B    |
| #1       | 1                     | 1  | 1  | 1  | 1  | 1  | 5                       | 5  | 5  | 4  | 5  | 5  | 3'36"            | 1'45" | 4'14" | 1'08" | 2'54" | 1'03" |
| #2       | 1                     | 1  | 1  | 1  | 0  | 1  | 5                       | 5  | 4  | 5  | 4  | 3  | 1'17"            | 0'40" | 4'59" | 2'31" | 3'59" | 2'43" |
| #3       | 1                     | 1  | 1  | 1  | 1  | 1  | 5                       | 5  | 4  | 5  | 4  | 3  | 0'37"            | 1'19" | 4'27" | 2'22" | 7'45" | 7'04" |
| #4       | 1                     | 1  | 1  | 1  | 1  | 1  | 5                       | 5  | 5  | 5  | 5  | 5  | 2'30"            | 0'14" | 2'49" | 0'34" | 2'47" | 1'49" |

<sup>1</sup>**Decision:** 1 and 0 denote the review’s decision is correct and incorrect, respectively; <sup>2</sup>**Confidence:** 1, 2, 3, 4, 5 represent reviewer’s decision confidence from the least confident to the most confident; <sup>3</sup>*Grey background* denotes the current case is applying XGUARDIAN;

Table 12: User study reviewers’ CS2 skill level, where *5EPlay* [1] and *Perfect World* [39] are the platform (introduced in *Ethical Considerations*) ranks, *Premier* is the official rank. All invited reviewers are the former reviewers of CS:GO Overwatch Investigators [53].

| Reviewer | CS2 Skill Level                                  | Playtime (hrs) |
|----------|--------------------------------------------------|----------------|
| #1       | 5EPlay - A, Perfect World - S                    | 3,563          |
| #2       | 5EPlay - A, Perfect World - A, Premier - 18,491  | 2,020          |
| #3       | 5EPlay - A, Perfect World - S, Premier - 19,486  | 2,400          |
| #4       | 5EPlay - A+, Perfect World - S, Premier - 20,000 | 2,917          |



# EXPLAINER's Potential Bias Investigation

- Reviewers with XGuardian have higher decision accuracy.

Table 11: User study results of the EXPLAINER's potential bias questionnaire.

| Reviewer | Decision <sup>1</sup> |    |    |    |    |    | Confidence <sup>2</sup> |    |    |    |    |    | Time Consumption |       |       |       |       |       |
|----------|-----------------------|----|----|----|----|----|-------------------------|----|----|----|----|----|------------------|-------|-------|-------|-------|-------|
|          | 1A                    | 1B | 2A | 2B | 3A | 3B | 1A                      | 1B | 2A | 2B | 3A | 3B | 1A               | 1B    | 2A    | 2B    | 3A    | 3B    |
| #1       | 1                     | 1  | 1  | 1  | 1  | 1  | 5                       | 5  | 5  | 4  | 5  | 5  | 3'36"            | 1'45" | 4'14" | 1'08" | 2'54" | 1'03" |
| #2       | 1                     | 1  | 1  | 1  | 0  | 1  | 5                       | 5  | 4  | 5  | 4  | 3  | 1'17"            | 0'40" | 4'59" | 2'31" | 3'59" | 2'43" |
| #3       | 1                     | 1  | 1  | 1  | 1  | 1  | 5                       | 5  | 4  | 5  | 4  | 3  | 0'37"            | 1'19" | 4'27" | 2'22" | 7'45" | 7'04" |
| #4       | 1                     | 1  | 1  | 1  | 1  | 1  | 5                       | 5  | 5  | 5  | 5  | 5  | 2'30"            | 0'14" | 2'49" | 0'34" | 2'47" | 1'49" |

<sup>1</sup>**Decision:** 1 and 0 denote the review's decision is correct and incorrect, respectively; <sup>2</sup>**Confidence:** 1, 2, 3, 4, 5 represent reviewer's decision confidence from the least confident to the most confident; <sup>3</sup>*Grey background* denotes the current case is applying XGUARDIAN;

# EXPLAINER's Potential Bias Investigation

- XGuardian's false determination does not affect the correctness of reviewer's final decision.

Table 11: User study results of the EXPLAINER's potential bias questionnaire.

| Reviewer | Decision <sup>1</sup> |    |    |    |    |    | Confidence <sup>2</sup> |    |    |    |    |    | Time Consumption |       |       |       |       |       |
|----------|-----------------------|----|----|----|----|----|-------------------------|----|----|----|----|----|------------------|-------|-------|-------|-------|-------|
|          | 1A                    | 1B | 2A | 2B | 3A | 3B | 1A                      | 1B | 2A | 2B | 3A | 3B | 1A               | 1B    | 2A    | 2B    | 3A    | 3B    |
| #1       | 1                     | 1  | 1  | 1  | 1  | 1  | 5                       | 5  | 5  | 4  | 5  | 5  | 3'36"            | 1'45" | 4'14" | 1'08" | 2'54" | 1'03" |
| #2       | 1                     | 1  | 1  | 1  | 0  | 1  | 5                       | 5  | 4  | 5  | 4  | 3  | 1'17"            | 0'40" | 4'59" | 2'31" | 3'59" | 2'43" |
| #3       | 1                     | 1  | 1  | 1  | 1  | 1  | 5                       | 5  | 4  | 5  | 4  | 3  | 0'37"            | 1'19" | 4'27" | 2'22" | 7'45" | 7'04" |
| #4       | 1                     | 1  | 1  | 1  | 1  | 1  | 5                       | 5  | 5  | 5  | 5  | 5  | 2'30"            | 0'14" | 2'49" | 0'34" | 2'47" | 1'49" |

<sup>1</sup>**Decision:** 1 and 0 denote the review's decision is correct and incorrect, respectively; <sup>2</sup>**Confidence:** 1, 2, 3, 4, 5 represent reviewer's decision confidence from the least confident to the most confident; <sup>3</sup>*Grey background* denotes the current case is applying XGUARDIAN;

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|----------|-----------------------|----|----|----|----|----|-------------------------|----|----|----|----|----|------------------|-------|-------|-------|-------|-------|
|          | 1A                    | 1B | 2A | 2B | 3A | 3B | 1A                      | 1B | 2A | 2B | 3A | 3B | 1A               | 1B    | 2A    | 2B    | 3A    | 3B    |
| #1       | 1                     | 1  | 1  | 1  | 1  | 1  | 5                       | 5  | 5  | 4  | 5  | 5  | 3'36"            | 1'45" | 4'14" | 1'08" | 2'54" | 1'03" |
| #2       | 1                     | 1  | 1  | 1  | 0  | 1  | 5                       | 5  | 4  | 5  | 4  | 3  | 1'17"            | 0'40" | 4'59" | 2'31" | 3'59" | 2'43" |
| #3       | 1                     | 1  | 1  | 1  | 1  | 1  | 5                       | 5  | 4  | 5  | 4  | 3  | 0'37"            | 1'19" | 4'27" | 2'22" | 7'45" | 7'04" |
| #4       | 1                     | 1  | 1  | 1  | 1  | 1  | 5                       | 5  | 5  | 5  | 5  | 5  | 2'30"            | 0'14" | 2'49" | 0'34" | 2'47" | 1'49" |

<sup>1</sup>**Decision:** 1 and 0 denote the review's decision is correct and incorrect, respectively; <sup>2</sup>**Confidence:** 1, 2, 3, 4, 5 represent reviewer's decision confidence from the least confident to the most confident; <sup>3</sup>*Grey background* denotes the current case is applying XGUARDIAN;

# EXPLAINER's Potential Bias Investigation

- XGuardian's false determination does not affect the correctness of reviewer's final decision.

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| Reviewer | Decision <sup>1</sup> |    |    |    |    |    | Confidence <sup>2</sup> |    |    |    |    |    | Time Consumption |       |       |       |       |       |
|----------|-----------------------|----|----|----|----|----|-------------------------|----|----|----|----|----|------------------|-------|-------|-------|-------|-------|
|          | 1A                    | 1B | 2A | 2B | 3A | 3B | 1A                      | 1B | 2A | 2B | 3A | 3B | 1A               | 1B    | 2A    | 2B    | 3A    | 3B    |
| #1       | 1                     | 1  | 1  | 1  | 1  | 1  | 5                       | 5  | 5  | 4  | 5  | 5  | 3'36"            | 1'45" | 4'14" | 1'08" | 2'54" | 1'03" |
| #2       | 1                     | 1  | 1  | 1  | 0  | 1  | 5                       | 5  | 4  | 5  | 4  | 3  | 1'17"            | 0'40" | 4'59" | 2'31" | 3'59" | 2'43" |
| #3       | 1                     | 1  | 1  | 1  | 1  | 1  | 5                       | 5  | 4  | 5  | 4  | 3  | 0'37"            | 1'19" | 4'27" | 2'22" | 7'45" | 7'04" |
| #4       | 1                     | 1  | 1  | 1  | 1  | 1  | 5                       | 5  | 5  | 5  | 5  | 5  | 2'30"            | 0'14" | 2'49" | 0'34" | 2'47" | 1'49" |

<sup>1</sup>**Decision:** 1 and 0 denote the review's decision is correct and incorrect, respectively; <sup>2</sup>**Confidence:** 1, 2, 3, 4, 5 represent reviewer's decision confidence from the least confident to the most confident; <sup>3</sup>*Grey background* denotes the current case is applying XGUARDIAN;

# Data collection generalizability

- Certain niche titles or those developed by smaller studios may lack replay system
- Developers can alternatively collect **pitch** and **yaw** data which are ALL FPS games must-haves

# Undetectable cases

- Cannot detect players who do not secure kills
  - e.g., purely support-oriented heroes
- Cannot detect a cheater employs techniques that do not influence the aiming trajectory
  - e.g., illegally sharing information with teammates

# XGuardian: Towards Generalized, Explainable and More Effective Server-side Anti-cheat in First-Person Shooter Games

Jiayi Zhang, Chenxin Sun, Chenxiong Qian

*HKU*

## Links

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Artifact



Paper



Demo